Cyclotomic Units, Volkenborn Integration, and p-adic L-functions

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1 Introduction

1.1 The p-adic numbers

Let p be a prime number. Given a rational number $r \in \mathbb{Q} \setminus \{0\}$, we may always uniquely write

$$r = \frac{a}{b} p^e$$

such that $p \nmid a$, $p \nmid b$, $\gcd(a, b) = 1$, and $a > 0$ for integers a, b, e. We define the p-adic valuation to be this number e, and write $v_p(r) = e$. For example:

$$v_3(9) = 2 \qquad v_5\left(\frac{3}{10}\right) = -1 \qquad v_7(33) = 0.$$

With the p-adic valuation we can define a new absolute value function on the rational numbers, called the p-adic absolute value. Specifically, we let

$$|x|_p = p^{-v_p(x)} \qquad \text{and} \qquad |0|_p = 0.$$

This gives us a corresponding p-adic distance function, $d_p(x, y) = |x - y|_p$. This distance function satisfies the triangle inequality

$$d_p(x, z) \leq d_p(x, y) + d_p(y, z),$$

and in fact satisfies something stronger, called the ultrametric inequality:

$$d_p(x, y) \leq \max\{d_p(x, y), d_p(y, z)\}.$$

Like with the standard absolute value $|\cdot|$, the rational numbers are *not* cauchy complete under $|\cdot|_p$. If you complete them in the same way as one does in real analysis, you obtain the p-adic numbers $\mathbb{Q}_p$.

One way to think of p-adic numbers is via their *p-adic expansion.* To start, write integers in base p. For example, in base 3, $120_3 = 15_{10}$. Then p-adic numbers can be thought of as base p numbers with digits that go on infinitely to the left. For example

$$\ldots 340431024.31 \in \mathbb{Q}_5.$$

If a p-adic number has no digits after the decimal point[1] then we say it is a p-adic integer, and we denote the set of p-adic integers by $\mathbb{Z}_p$.

The reason the digits go to the left is because, under the p-adic absolute value, numbers like

$$10 \cdots 00$$

are small, whereas under the usual absolute value, numbers like

$$0.00 \cdots 001$$

are small, and digits should go in the direction that makes numbers smaller.

We can add p-adic numbers by using the addition algorithm from elementary school:

[1] Technically, since we aren't in base 10, its only called the radix point.

$$
\begin{array}{r}
1 \\
\cdots 1201 \\
+\cdots 1002 \\
\hline
\cdots 2210
\end{array}
$$

Where note that we carry using base-p arithmetic, so since $1+2=10$ in base 3, we carry the 1 after the first addition. You can also subtract, multiply, and divide them with similar methods, thus turning $\mathbb{Q}_p$ into a field. Finally, you can read the p-adic valuation from these expansions, its exactly the number of zeroes at the end:

$$v_7(\ldots 536400) = 2.$$

The process of Cauchy completion with respect to $|\cdot|_p$ gives us some new numbers, just like how Cauchy completion using the usual absolute value gives us irrational numbers. For example, even though $i \notin \mathbb{Q}$, we have

$$\sqrt{-1} = \ldots 013233 \in \mathbb{Q}_5.$$

so i is in the 5-adics. There's a connection here to modular arithmetic. Notice that

$$
\begin{aligned}
3^2 &\equiv -1 \bmod 5 \\
(3 \cdot 5 + 3)^2 &\equiv -1 \bmod 5^2 \\
(2 \cdot 5^2 + 3 \cdot 5 + 3)^2 &\equiv -1 \bmod 5^3 \\
&\vdots
\end{aligned}
$$

so we can think of every fact in the p-adic numbers as being a sort of infinite tower of facts about the integers mod powers of p.

Because the p-adic numbers are complete with respect to the p-adic absolute value, we can do analysis on them, with the standard definitions of limits and continuity extending word-for-word. For example, if $f : \mathbb{Q}_p \to \mathbb{Q}_p$ is a function, then we can say f is continuous at x if for all $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|x-y|_p < \delta \implies |f(x) - f(y)|_p < \varepsilon.$$

Notice how, even though we are working with p-adic functions, the numbers ε and δ are still real numbers.

We can also differentiate and integrate functions from p-adics to p-adics, but this is a lot more subtle.

Finally, there is also a p-adic version of the complex numbers, $\mathbb{C}_p$, though unlike the reals you do not get to it by simply adding in a square root of -1.

1.2 L-functions and p-adic L-functions

Let m be a positive integer. A Dirichlet character mod m is a function $\chi : \mathbb{Z} \to \mathbb{C}$ that satisfies the following properties:

$$
\begin{aligned}
\chi(ab) &= \chi(a)\chi(b) \\
\chi(a+m) &= \chi(a) \\
\chi(a) &= 0 \quad \text{if} \quad \gcd(m,a) > 1.
\end{aligned}
$$

Notice for example the constant function $\chi_0(n) = 1$ is a Dirichlet character. Associated to each Dirichlet character, you obtain a Dirichlet L-function $L(\chi, s)$

$$L(\chi, s) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

The special case of $\chi = \chi_0$ gives you the Rieman zeta function

$$L(\chi_0, s) = \zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

The Dirichlet L-functions are important in number theory since their behavior seems to govern the behavior of number-theoretic questions. For example, the zeroes of the Riemann zeta function are known to encode information about the distribution of prime numbers. However, that particular story is not what we are interested in today. For each positive integer n and Dirichlet character χ, there is a number $B_{n,\chi}$ called the nth χ-Bernoulli number. These numbers encode a lot of important arithmetic information in algebraic number theory, and one can prove that for every $n > 0$

$$L(\chi, 1-n) = \frac{B_{n,\chi}}{n}.$$

In 1851, Kummer [Kum51] noticed that some Bernoulli numbers satisfy certain modular congruences. Specifically, if h, k are even numbers not divisible by $p-1$ and $h \equiv k \bmod (p-1)p^a$

$$(1-p^{k-1})\frac{B_k}{k} \equiv (1-p^{h-1})\frac{B_h}{h} \bmod p^a.$$

Congruences like these are usually the sign of something p-adic happening in the background, because two numbers that are congruent modulo a high power of p are p-adically close to each other. However, Kummer did not realize this at the time since p-adic numbers had not been invented yet. Instead, this was only truly realized by Kubota and Leopoldt in the 1960s. They realized the Kummer congruences suggested they arose from a certain continuous p-adic function. To state their result, we need something called the Teichmüller character, written as ω. This is a specific Dirichlet character mod p whose definition is a bit involved, so we omit it. The rough idea is that the value of $\omega(a)$, when viewed as a p-adic number, should be congruent mod p to a. For example, since $i \equiv 3 \bmod 5$ (see where we explained how $i \in \mathbb{Z}_5$), we have $\omega(3) = i$.

Theorem 1.1 ([KL64]). *There exists a p-adically continuous function $L_p(\chi, s)$ such that for all positive integers n*

$$L_p(\chi, 1-n) = (1 - \chi\omega^{-n}(p)p^{n-1})\frac{B_{n,\chi\omega^{-n}}}{n}.$$

The function $L_p(\chi, s)$ is called the p-adic Dirichlet L-function of χ.

Like regular L-functions, the p-adic L-functions encode number-theoretic information within them. However, their information is much more subtle, and involves something called the Cyclotomic $\mathbb{Z}_p$-extension, which will be explained in section 1.4.

There is one more important thing to note about p-adic L-functions, which is that they can be thought of as certain power series. To explain this, we will need a bit more knowledge about Dirichlet characters.

Specifically, let χ be a character mod m that we want to study the p-adic L-function of. Each character χ comes with a certain decomposition, $\chi(n) = \rho(n)\psi(n)$. The character ρ is called the "tame part", and the character ψ is called the "wild part". If $m = dp^r$, with $\gcd(d, p) = 1$, then ρ is a character mod d or dp and ψ is a character mod p^r. Finally, we require that $\psi(n)^{p^r} = 1$ if $\gcd(n, p) = 1$. These properties turn out to exactly characterize ρ and ψ, so this decomposition is unique.

Now let χ be a character mod d or dp with $p \nmid d$. In such a case, the wild character in the decomposition from the previous paragraph is trivial, so we call this character tame. On the other hand, if ψ is a character mod p^r, we say it is purely wild if the tame character in its decomposition is trivial. Iwasawa showed in [Iwa69] that unless $\chi = \chi_0$, then there is a unique power series $F_\chi(T)$ with coefficients in $\mathbb{C}_p$ (we can say something more precise than $\mathbb{C}_p$ here, but it is unnecessary for now) such that for all wild characters ψ and all p-adic integers s we have

$$L_p(\chi\psi, s) = F_\chi(\psi(1+pd)(1+pd)^s - 1).$$

There are a couple really interesting observations to make here. First, the role of the tame part χ and the wild part ψ is fundamentally different. While the tame part determines the power series F_χ, the tame part only contributes a minor factor to the formula. Secondly, the fact that the p-adic L-function is essentially encoded as a power series is quite remarkable. While arbitrary smooth functions are hard to work with, power series are much more rigid, and that will enable many of the remarkable results we discuss in the forthcoming sections.

1.3 The contribution of this thesis

The main result of this thesis is a new formula for p-adic L-functions. Let χ be a Dirichlet character mod p with $\chi(-1) = 1$. We will generalize it further later. We will define two sequences of p-adic numbers, $u_\chi = (u_{\chi,n})$ and $c_\chi = (c_{\chi,n})$. These sequences will come from a $\mathbb{Z}_p$-extension (an object explained in the next section,) but we do not need to know that for now.

We can explicitly write down elements $(c_{\chi,n})$, but it is quite involved, so we will satisfy ourselves with just $c_{\chi,1}$. Let $\zeta_p \in \mathbb{C}_p$ be a fixed p-adic number such that $\zeta_p^p = 1$ but $\zeta_p \neq 1$. Then

$$c_{\chi,1} = \prod_{a=1}^{p-1} (1 - \zeta_p)^{\chi(a)}.$$

The other sequence, $(u_{\chi,n})$, can also kind of be written down, but its a bit more subtle. We can definitely write $u_{\chi,1}$:

$$u_{\chi,1} = \exp_p \left(\sum_{a=1}^{p-1} \zeta_p^a \chi(a) \right)$$

where $\exp_p$ is a p-adic version of the exponential function $x \mapsto e^x$.

For $u_{\chi,n}$ for larger n, we know what the logarithm $\log_p(u_{\chi,n})$ should be ($\log_p$ is the p-adic natural logarithm, not base p-logarithm.) It has an explicit formula in terms of the base p-digits of ζ_{p-1}. What we would like to do is simply take $\exp_p$ of what its logarithm should be and obtain $u_{\chi,n}$. However, the radius of convergence of $\exp_p$ is *too small* to use it for this purpose. Instead, we will have to take a significant detour through algebraic number theory to prove that u_χ exists. (By "exists" we mean that there is a sequence of p-adic numbers whose logarithm looks like what we expect.) The unconditional existence of u_χ is progress on [AW17], who were only able to prove that u_χ exists under a technical assumption on p.

The sequences u_χ and c_χ have a special property called norm coherence, which is a condition from Galois theory. We define a function ϱ that takes a norm coherent sequence $\alpha = (\alpha_n)$ to something called a "Volkenborn distribution," which is a p-adic analytic object considered first in [Bar02]. The key thing you can do with Volkenborn distributions is integrate over the p-adic numbers with them. The best way to explain this involved measure theory, but intuitively we can think of a Volkenborn distribution as being a generalized dx. This allows us to define the function

$$f_\alpha(T) = \int_{\mathbb{Z}_p} (T+1)^x d\varrho(\alpha).$$

Now, the information we know about the logarithm of $u_{\chi,n}$ and the explicit formula for $c_{\chi,n}$ lets us prove a following formula for p-adic L-functions. If ψ is a purely wild character, then

$$\frac{f_{c_\chi}(\zeta_\psi - 1)}{f_{u_\chi}(\zeta_\psi - 1)} = L_p(\chi\psi, 1).$$

The numbers ζ_ψ are p-adic numbers that can be determined uniquely by the character ψ.

Strictly speaking, the order I've explained this in is slightly backwards. The function ϱ is a crucial piece of proving that u_χ exists. What we actually do is start with what f_{u_χ} "should" be, and work backwards to determine what the logarithm of u_χ should be, and then use deep facts from algebraic number theory to show that u_χ actually exists.

Now, what we do in this paper is slightly more general than what has been presented so far. Instead of starting with a mod p character χ, we want to be able to start with an arbitrary character. This adds quite a bit of technical difficulty. There are roughly three reasons, that are a bit difficult to explain. One comes from algebraic number theory, and two come from representation theory. The issue from algebraic number theory is relatively easy to dispose of by making a minor adjustment in certain cases.

The issues from representation theory are harder to deal with, and easier to explain, so we go in a bit more depth on them. The first comes from the fact that in the previous case considered, $\chi(a) \in \mathbb{Z}_p$ in all cases. However with our broader set of characters being considered now, $\chi(a)$ might only be in $\mathbb{C}_p$. This makes some of our algebraic calculations harder to do.

The second issue from representation theory is even harder to deal with. Let φ be Euler's totient function, which takes n to the order of the group $(\mathbb{Z}/n\mathbb{Z})^\times$. Now let χ be a character mod m. We say we are in the *semisimple case* if $p \nmid \varphi(m)$, and we are in the *non-semisimple case* if $p \mid \varphi(m)$. The non-semisimple case is significantly harder than the semisimple case, to the point where we cannot handle it in full generality. Even just handling the special case where $m = q^r p$ is the product of a prime power q^r and p requires significant work.

1.4 $\mathbb{Z}_p$ extensions and Iwasawa theory

The following section assumes the reader has taken abstract algebra.

In algebraic number theory, we are interested in certain subsets of $\mathbb{C}$ called *number fields*. A number field $K \subseteq \mathbb{C}$ has the following properties.

1. K is a field: it contains 0 and 1, and you can add, subtract, multiply, and divide by nonzero elements of K.

2. K is finite dimensional as a vector space over $\mathbb{Q}$.

To explain what we mean by (2.), notice that since $1 \in K$, and we can add and subtract in K, we see that $\mathbb{Z} \subseteq K$. Then, since we can also divide, it follows that $\mathbb{Q} \in K$. This then means K satisfies the vector space axioms for scalars in $\mathbb{Q}$. This finite dimensionality constraint essentially means that there are a finite list of elements $\alpha_1, \ldots, \alpha_n \in K$ such that

$$K = \{a_1\alpha_1 + \cdots + a_n\alpha_n : a_i \in \mathbb{Q}\}.$$

For example, this means that even though $\mathbb{C}$ satisfies condition (1.), it does not satisfy condition (2.), so it is not a number field.

The simplest way to create number fields is by adjoining elements to $\mathbb{Q}$. Here's what that means precisely. Let $\alpha \in \mathbb{C}$ be a complex number such that there is a polynomial $p(x)$ of degree n with rational coefficients such that $p(\alpha) = 0$. Such an α is called algebraic, and the set

$$\mathbb{Q}(\alpha) = \{y_0\alpha^0 + y_1\alpha + \cdots + y_{n-1}\alpha^{n-1} : y_i \in \mathbb{Q}\}$$

forms a number field. The α_i in the finite dimensionality condition can be α^{i-1}, and the sum or difference of two elements of K are in K. For multiplication, you can distribution the product and then use the identity $p(\alpha) = 0$ to simplify back into something of the form $y_0\alpha^0 + \cdots + a_{n-1}\alpha^{n-1}$. The hardest thing to show is that you can divide, which we omit. The proof can be found in a book on field theory. We can give in example, which explains how this is essentially a generalization of rationalizing the denominator. Say we have the extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ and want to show the fraction $\frac{3+2\sqrt{2}}{1-5\sqrt{2}}$ can be written in the form $a + b\sqrt{2}$, then we have

$$\frac{3+2\sqrt{2}}{1-5\sqrt{2}} = \frac{(3+2\sqrt{2})(1+5\sqrt{2})}{(1-5\sqrt{2})(1+5\sqrt{2})} = \frac{23+17\sqrt{2}}{1-50} = \frac{-23}{49} + \frac{-17}{49}\sqrt{2}.$$

Now we will give an example of a cyclotomic $\mathbb{Z}_p$-extension, which is a certain "tower" of number fields. Specifically, consider the number fields

$$K_n = \mathbb{Q}(e^{2\pi i/p^{n+1}}).$$

These number fields have a containment relation: $K_n \subseteq K_{n+1}$. We call the "tower" of number fields $K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$ the **cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}(e^{2\pi i/p})$.**

Each of these numbers fields K_n has a finite abelian group C_n associated with it, called its ideal class group. The definition of C_n is beyond the scope of this introduction, but all that matters is that number theorists are interested in the structure and size of C_n.[2]

Now, it cannot be understated how difficult of a problem computing C_n is. Even finding the class group of a specific number field is incredibly hard, so to simplify, we ask if we can determine the Sylow-p subgroup

[2] Historically, we were interested in the size of C_n because Kummer found a proof of Fermat's last theorem for primes p where $p \nmid C_p$.

A_n of C_n, which is hopefully an easier problem. The early history of Iwasawa theory is the story of how we determined the structure of A_n. First, in [Iwa58, Iwa59], the author (and the namesake of the field) proved that there are constants λ, μ, ν such that for "big enough" n we have

$$|A_n| = p^{\lambda n + \mu p^n + \nu}.$$

Now let's assume that $p > 2$. Then we can split A_n into two subgroups, called the "plus part" and the "minus part" where:

$$A_n = A_n^+ \oplus A_n^-.$$

Iwasawa's methods also gave us numbers $\lambda^\pm$, $\mu^\pm$, and $\nu^\pm$ such that

$$|A_n^+| = p^{\lambda^+ n + \mu p^n + \nu^+} \qquad \text{and} \qquad |A_n^-| = p^{\lambda^- n + \mu^- p^n + \nu^-}$$

The behavior of A_n^+ and A_n^- are wildly different, or at least are conjectured (with significant evidence) to be wildly different.

For A_n^+ there are two well-known conjectures in algebraic number theory, Vandiver's conjecture and Greenberg's conjecture[Gre71], that would both imply that A_n^+ is the trivial group. This would of course imply that $\lambda^+ = \mu^+ = \nu^+ = 0$. However, both of these conjectures are wide open.

On the other hand, A_n^- is typically highly nontrivial, and its structure is intimately related to the p-adic L-functions $L_p(\chi, s)$. Let χ be a character mod p such that $\chi(-1) = 1$. There are exactly $(p-1)/2$ of these. The power series $F_\chi(T)$ discussed in the previous section have coefficients not just in $\mathbb{C}_p$, but actually $\mathbb{Z}_p$.

We can then define two numbers, $\mu(F_\chi)$ and $\lambda(F_\chi)$. To compute $\mu(F_\chi)$, determine the exponent of the largest power of p that divides all the coefficients of F_χ. So that means $F_\chi(T)/p^{\mu(F_\chi(T))}$ will still have coefficients in $\mathbb{Z}_p$ (you won't introduce any denominators) but one of the coefficients will not be divisible by p. Next, count the number of zeroes of $F_\chi(T)$. We can prove that this number will be finite, and we call that number $\lambda(F_\chi(T))$. We can show

$$\lambda^- = \sum_\chi \lambda(F_\chi) \qquad \text{and} \qquad \mu^- = \sum_\chi \mu(F_\chi)$$

where the sums are taken over those χ with period p and $\chi(-1) = 1$. This is our first glimpse at the *Iwasawa Main Conjecture*, which posits a deep and precise relationship between A_n and $L_p(\chi, s)$. It was first conjectured by Ralph Greenberg in [Gre75], and (despite its name) was proven by Barry Mazur and Andrew Wiles in [MW84]. We will attempt to describe what the main conjecture is. This is difficult and will require us to assume a certain level of fluency with abstract algebra, but we hope the reader can get the ideas.

To explain the main conjecture, we will need to introduce further structure on A_n. They are what we call Galois modules. This is an additional structure on top of being an abelian group, where the group $(\mathbb{Z}/p^{n+1}\mathbb{Z})^\times$ has a natural action on A_n. Now, there is an important isomorphism

$$(\mathbb{Z}/p^{n+1}\mathbb{Z})^\times \cong (\mathbb{Z}/p\mathbb{Z})^\times \times (\mathbb{Z}/p^n\mathbb{Z}). \tag{1}$$

We can use the first factor, $(\mathbb{Z}/p\mathbb{Z})^\times$, to decompose A_n into $p-1$ pieces (note that the cardinality of $(\mathbb{Z}/p\mathbb{Z})^\times$ is $p-1$), and we write this decomposition

$$A_n \cong A_n^{(0)} \oplus A_n^{(1)} \oplus \cdots \oplus A_n^{(p-2)}$$

This decomposition uses up the action of $(\mathbb{Z}/p\mathbb{Z})^\times$, but we will retain the action of $(\mathbb{Z}/p^n\mathbb{Z})$ on each $A_n^{(i)}$. This decomposition is compatible with the $+, -$ decomposition in the sense that

$$A_n^+ = A_n^{(0)} \oplus A_n^{(2)} \oplus \cdots \oplus A_n^{(p-3)} \qquad \text{and} \qquad A_n^- = A_n^{(1)} \oplus A_n^{(3)} \oplus \cdots \oplus A_n^{(p-2)}.$$

Now we will vigorously wave our hands. There is a way to "combine" all the A_ns together, into an object called X_∞, which we refer to as the "unramified Iwasawa module." At the same time, the actions of $(\mathbb{Z}/p^{n+1}\mathbb{Z})^\times$ "combine" to give an action of $\mathbb{Z}_p^\times$ on X_∞. Now, the analogue of 1 in this case is an isomorphism $\mathbb{Z}_p^\times \cong (\mathbb{Z}/p\mathbb{Z})^\times \times \mathbb{Z}_p$. We once again use the $(\mathbb{Z}/p\mathbb{Z})^\times$ action to decompose X_∞ into pieces:

$$X_\infty \cong X_\infty^{(0)} \oplus X_\infty^{(1)} \oplus \cdots \oplus X_\infty^{(p-2)}.$$

We also have a $+$ and $-$ decomposition, like with A_n, and the decompositions are compatible in the same way.

Now, we are left with these abelian groups $X_\infty^{(i)}$, which somehow combine the $A_n^{(i)}$ for all n, and they have a group action of $\mathbb{Z}_p$ on them. It turns out that having a group action of $\mathbb{Z}_p$ implies they are "modules" over the ring

$$\Lambda = \mathbb{Z}_p[\![T]\!] = \{f(T) = a_0 + a_1 T + \cdots : a_i \in \mathbb{Z}_p\}.$$

This is the ring of formal power series with coefficients in the p-adic integers. What it means for $X_\infty^{(i)}$ to be a module over Λ is that given $f \in \Lambda$ and $x \in X_\infty^{(i)}$, we can "multiply" x by f and get an element $f \cdot x \in X_\infty^{(i)}$. We call any abelian group with a multiplication by elements of Λ a Λ-module.

Now, given any (sufficiently small in a precise sense) Λ-module Y, we can define something called the characteristic ideal

$$\mathrm{Ch}_\Lambda(Y) \subseteq \Lambda$$

This is a subset of Λ which consist of the multiples of some polynomial $g \in \Lambda$. There is some choice in what g is, but the set of its multiples is always fixed.

For each $i \in \{0, 1, \cdots, p-2\}$, there is a corresponding character χ_i with period p (or 1.) We can now state the main conjecture:

Theorem 1.2 (The main conjecture of Iwasawa theory)**.** *Let $i \in \{1, 3, \cdots, p-2\}$. Then there is an equality of sets*

$$Ch_\Lambda(X_\infty^{(i)}) = \{f \in \Lambda : f \text{ is a multiple of } F_{\chi_i}(T)\}.$$

Now, here's what this means: the set $\mathrm{Ch}_\Lambda(X_\infty^{(i)})$ was an object defined purely with abstract algebra, no analysis in sight. On the other hand, $F_\chi(T)$ is essentially the same information as the p-adic L-function $L_p(\chi, s)$, which is defined purely with analysis. So this theorem tells us that these two worlds communicate exactly.

Now, recall the discussion of A_n^+ from before, where it was conjectured but not proven that it was always trivial: that would follow from Vandiver's conjecture. Suppose this conjecture is true (in practice, we can verify it for any prime number p.) In this case, we have an even better description of X_∞. For $f \in \Lambda$, define the *quotient ring*

$$\Lambda/f\Lambda = \{g + f\Lambda : g \in \Lambda\}.$$

This has the structure of a Λ module. The following is essentially a special case of Theorem 1.2,[3] and was proven by Iwasawa in [Iwa69].

Theorem 1.3. *Assume Vandiver's conjecture holds for p. Then for odd i we have an isomorphism of Λ-modules*

$$X_\infty^{(i)} \cong \Lambda/F_{\chi_i}(T)\Lambda,$$

and for even i we have

$$X_\infty^{(i)} \cong 0.$$

It turns out that we can recover the abelian group structure of $A_n^{(i)}$ from the Λ-module structure of $X_\infty^{(i)}$, so Theorem 1.3 is the answer to the question that began this whole quest: what is the group structure of A_n?

[3]Technically, this result is quite a bit easier to prove than Theorem 1.2, but it is morally quite similar.

1.5 Technical Introduction

The following section assumes the reader is familiar with Iwasawa theory and is an introduction to the specific contents of this thesis. For the remainder of this thesis we assume p is an odd prime. Our result will end up being parametrized by a p-free integer m', but we will state the case for $m' = 1$ for now. Define $K = \mathbb{Q}_p(\zeta_p)$ and let K_∞/K denote the cyclotomic $\mathbb{Z}_p$ extension of $K = K_0$. Let $\check{K}_n^\times$ denote the p-completion of the multiplicative group of K_n, and $A(K_\infty) = \varprojlim \check{K}_n^\times$ be their inverse limit. Let $\Gamma = \mathrm{Gal}(K_\infty/K)$ and $\Delta = \mathrm{Gal}(K/\mathbb{Q}_p)$, and $\Lambda = \mathbb{Z}_p[\![\Gamma]\!]$. Then we have an isomorphism of $\Lambda[\Delta]$ modules[NSW08, Prop. 11.2.3]

$$A(K_\infty) \cong \Lambda[\Delta] \oplus \mathbb{Z}_p(1).$$

where $\mathbb{Z}_p(1) = \varprojlim \mu_{p^n}$ is the Tate twist of $\mathbb{Z}_p$.

Let $\omega : \Delta \to \mathbb{Z}_p^\times$ be the Teichmuller character, which is the unique section of the projection morphism $\mathbb{Z}_p^\times \to (\mathbb{Z}/p\mathbb{Z})^\times \cong \Delta$. We define the *idempotents* ε_i for $i \in \{0. \dots, p-2\}$ by

$$\varepsilon_i = \frac{1}{p-1} \sum_{\delta \in \Delta} \omega^{-i}(\delta)[\delta] \in \mathbb{Z}_p[\Delta].$$

Given a $\mathbb{Z}_p[\Delta]$-module M, the ε_i act as projectors onto the eigenspace of M where Δ acts through ω^i. The isomorphism $A(K_\infty) \cong \Lambda[\Delta] \oplus \mathbb{Z}_p(1)$ then splits up under the idempotents

$$\varepsilon_i A(K_\infty) \cong \begin{cases} \Lambda & i \neq 1 \\ \Lambda \oplus \mathbb{Z}_p(1) & i = 1 \end{cases}.$$

Let $\gamma \in \Gamma$ be such that $\gamma \zeta_{p^n} = \zeta_{p^n}^{1+p}$. Selecting this generator amounts to fixing an isomorphism $\Gamma \cong \mathbb{Z}_p$. The space of K_∞-valued distributions on $\mathbb{Z}_p$ ($\mathcal{D}(\mathbb{Z}_p, K_\infty)$) is then isomorphic to the group ring $K_\infty[\![\Gamma]\!]$. We note the following subtltey: there are two natural choices of Γ-action on $K_\infty[\![\Gamma]\!]$. For $\gamma_0 \in \Gamma$, we can have either

$$\gamma_0 \left(\sum_{\gamma_1} \alpha_{\gamma_1}[\gamma_1] \right) = \sum_{\gamma_1} \gamma_0(\alpha_{\gamma_1})[\gamma_1]$$

or

$$\gamma_0 \left(\sum_{\gamma_1} \alpha_{\gamma_1}[\gamma_1] \right) = \sum_{\gamma_1} \alpha_{\gamma_1}[\gamma_0\gamma_1].$$

We are using the second convention.

As mentioned before, we have an isomorphism $K_\infty[\![\Gamma]\!] \cong \mathcal{D}(\mathbb{Z}_p, K_\infty)$. We will define a map $\varrho : A(K_\infty) \to \mathcal{D}(\mathbb{Z}_p, K_\infty)$ as follows: given $x = (x_n) \in A(K_\infty)$, define

$$\varrho(x)(a + p^n\mathbb{Z}_p) = \mathrm{Log}_p(x_n^{\gamma^a}).$$

where Log_p is the Iwasawa logarithm, an extension of $\log_p(1+T)$ to the entirety of $\mathbb{C}_p$. Then ϱ is a homomorphism of Λ-modules, and one can compute $\ker(\varrho) = \mathbb{Z}_p(1)$.

Consider the element $c = ((1 - \zeta_{p^{n+1}}))_{n \in \mathbb{N}} \in A(K_\infty)$. Its $\Lambda[\Delta]$ span defines a submodule $C_\infty^* \subseteq A(K_\infty)$. Let χ be a nontrivial even character of Δ. We can show that $c_\chi := (p-1)\varepsilon_\chi c$ generates $\varepsilon_\chi C_\infty^*$ because $(p-1)$ is invertible and c generates C_∞^*. Then

$$\varrho(c_\chi)(1 + p^n\mathbb{Z}_p) = \sum_{\delta \in \Delta} \overline{\chi}(\delta) \log_p(1 - \delta(\zeta_{p^{n+1}})),$$

if we select $n = 0$, then we obtain the formula

$$\varrho(c_\chi)(\mathbb{Z}_p) = \sum_{a=0}^{p-1} \overline{\chi}(a) \log_p(1 - \zeta_p^a),$$

where $\overline{\chi}$ is the inverse character of χ. This appears in the special value formula for $L_p(\chi, 1)$[Was12, Thm. 5.18]. This generalizes as follows: The function $a \mapsto \zeta_{p^n}^a$ is a locally constant function, and can thus be integrated against any distribution, including the unbounded distribution $\varrho(c_\chi)$. Let ψ be an arbitrary purely wild character of conductor p^{n+1} and let $\zeta_\psi = \psi(1+p)$. We then have

$$\int_{\mathbb{Z}_p} \zeta_\psi^x d\varrho(c_\chi)(x) = \sum_{a=0}^{p^{n+1}} \overline{\chi}\psi(a)\log_p(1-\zeta_{p^{n+1}}^a).$$

This is then part of the special value formula for $L_p(\chi\overline{\psi},1)$.

The main result of this paper, at least in the case $m=p$, is

Theorem 1.4 (Special case of Theorem 5.6(a))**.** *Let χ be a nontrivial even character of conductor p. Then there exists an element $u_\chi \in A(K_\infty)$ that generates $\varepsilon_\chi A(K_\infty)$ and obeys*

$$\varrho(u_\chi)(a+p^n\mathbb{Z}_p) = -\frac{1}{p^n}\sum_{b=0}^{p^n-1} \zeta_{p^n}^{-ab}\tau(\chi\psi_{\zeta_{p^n}^a})$$

where $\psi_{\zeta_{p^n}^a}$ is the unique purely wild character with $\psi_{p^n}^a(1+p) = \zeta_{p^n}^a$.

As a consequence, we obtain the formula

$$\frac{\int_{\mathbb{Z}_p} \zeta_{\overline{\psi}}^x d\varrho(c_\chi)(x)}{\int_{\mathbb{Z}_p} \zeta_{\overline{\psi}}^x d\varrho(u_\chi)(x)} = L_p(\chi\psi,1). \tag{2}$$

We will explain how this is related to a result of Iwasawa in [Iwa64]. Let U_n be the units of K_n, and $U_\infty = \varprojlim U_n$ under the norm maps. U_∞ and $A(K_\infty)$ are very closely related: if χ is not the trivial character, then

$$\varepsilon_\chi U_\infty = \varepsilon_\chi A(K_\infty)$$

so we may work with U_∞ instead of $A(K_\infty)$ for the sake of Theorem 1.4. Let $C_\infty = C_\infty^* \cap U_\infty$, and we similarly obtain $\varepsilon_\chi C_\infty^* = \varepsilon_\chi C_\infty$ for nontrivial χ. Then Iwasawa showed that for (even nontrivial) χ we have

$$\varepsilon_\chi(U_\infty/C_\infty) \cong \Lambda/G_\chi(T)$$

where $G_\chi(T)$ is the power series related to the p-adic L-functions, described in 1.2. Theorem 1.4, and espetially the corollary 2, is essentially a reinterpretation of Iwasawa's result in different, more explicit, language.

We now explain the generalization we study. Firstly, instead of working purely locally, we work semilocally. More precisely, let m' be a p-free integer, $m = m'p$, $F = \mathbb{Q}(\zeta_{m'})$, and $k = \mathbb{Q}(\zeta_m)$. Let k_∞/k be the cyclotomic $\mathbb{Z}_p$ extension of $\mathbb{Q}$, and define $K_n = k_n \otimes_{\mathbb{Q}} \mathbb{Q}_p$. If L_n is the localization of k_n at a prime above p, then $L_n = K_n^r$, where r is the number of primes p splits into in any of F, k_n, k_∞.

We can define $A(K_\infty)$, C_∞^*, U_∞, and C_∞ the same as before, just now in this semilocal setting. Gillard [Gil79a, Gil79b] generalized the isomorphism $\varepsilon_\chi U_\infty/C_\infty \cong \Lambda/G_\chi(T)$ to this setting. Assume $p \nmid \varphi(m)$ (where φ is Euler's totient function), and let χ be a character of $\Delta := \mathrm{Gal}(k/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$, then we have the following idempotent elements

$$\varepsilon_\chi = \frac{1}{|\Delta|}\sum_{\delta\in\Delta} \overline{\chi}(\delta)[\delta] \in \mathbb{Z}_p[\chi(\delta) : \delta \in \Delta][\Delta].$$

For convenience, we refer to the ring $\mathbb{Z}_p[\chi(\delta) : \delta \in \Delta]$ as $\mathbb{Z}_p[\chi]$. We note that ε_χ is not in $\mathbb{Z}_p[\Delta]$: this means we will have to perform a change of basis to $\mathbb{Z}_p[\chi]$ to meaningfully multiply by ε_χ. Gillard showed that in most cases with an even nontrivial character χ that

$$\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes (U_\infty/C_\infty)) \cong \Lambda[\chi]/G_\chi(T).$$

In the "nongeneric" case, then we don't quite an isomorphism but we do have a related exact sequence. Using this, we are able to obtain the entirety of Theorem 5.6.

Now, we can the case where $p \nmid \varphi(m)$ the "semisimple" because the expressions

$$\varepsilon_\chi = \frac{1}{|\Delta|} \sum_{\delta \in \Delta} \overline{\chi}(\delta)[\delta]$$

land in an integral extension of $\mathbb{Z}_p[\Delta]$. But when $p | \varphi(m)$, then the division by $|\Delta|$ means we only land in $\mathbb{Q}_p[\chi][\Delta]$. The lack of idempotents cause a major issue, and instead we have to work with the "χ-part" of a module M, denoted M^χ, which is intuitively the maximal submodule of M where Δ acts through χ. This object is significantly more poorly behaved that ε_χ, and homological constructions enter the picture.

[Gre92] constructs an analogue of Iwasawa's theorem in this case by tensoring every object involved with $\mathbb{Q}_p$, essentially adding denominators. This gets around the homological problems since the category of $\mathbb{Q}_p[\Delta]$-modules is semisimple. However, this does not suffice in our case, as we want u_χ to be an actual element of $A(K_\infty)$, not an element of $\mathbb{Q}_p \otimes_{\mathbb{Z}_p} A(K_\infty)$. However, Greither conjectured that an integral version of his results hold, and indeed they do: in [Tsu99] the author provides an honest integral version of Greither's results.

Just Tsuji's results are not enough to obtain the analogue of 1.4, since every part of its proof takes advantage of Idempotents, not just the contribution from the structure of $\varepsilon_\chi(U_\infty/C_\infty)$. Because of this, we will have to pay special attention to the Galois module structure of C_∞^* and compute some Ext groups involving it. As a consequence of this, we are only able to obtain results for $m = q^r p$ and χ satisfying a technical condition, rather than arbitrary m divisible only once by p, and arbitrary χ.

We do not believe that this is because the theorem is false in other cases, instead, the homological constructions in the proof will get highly complex as the Galois module structure of C_∞^* isn't very nice. In particular, the proof of analogues of 6.4 in those cases become much more difficult.

After the construction of u_χ, we give two applications. In the first, we describe the value of $L_p(\chi\psi, 1)$ as a sort of "explicit index formula." We can view the map ϱ as essentially a version of Log_p defined on $A(K_\infty)$ that preserves the Λ-module structure. Taking Γ-coinvariance, it literally is a logarithm, and we see that

$$L_p(\chi, 1) = \frac{\mathrm{Log}_p(c_{0,\chi})}{\mathrm{Log}_p(u_{0,\chi})},$$

so in this sense we have a type of 1D regulator for U_0/C_0. When adding a wild character ψ, we can think of the formula as saying

$$L_p(\chi\psi, 1) = \frac{\mathrm{Log}_p(c_{\psi,\chi})}{\mathrm{Log}_p(u_{\psi,\chi})}.$$

for "ψ-parts" $c_{\psi,\chi}$ and $u_{\psi,\chi}$ of c_χ and u_χ. These do not actually exist as units, but their logarithms do exist in some meaningful sense, and this allows us to intepret $L_p(\chi\psi, 1)$ similarly as a 1D regular, this time for the ψ-part of U_n/C_n, where ψ has order p^n. This application is explained in the discussion following 5.7.

Another application we give is to the image of Log_p on $\check{K}_n^\times$. Since we have generators u_χ of parts of $A(K_\infty)$, we may take their Γ^{p^n} coinvariance and obtain generators $u_{n,\chi}$ of corresponding parts of $\check{K}_n^\times$. Since we know $\varrho(u_\chi)$, we know the p-adic logarithm of $u_{n,\chi}$, and can therefore understand the image of trace on the χ part of U_n. This is done in Proposition 5.9.

We note the following relationship to previous work. The relationship between c_χ and $L_p(\chi, 1)$ has been noted in the literature, for example in [BCD+14, Sec. 1.1]. We will explain what they have done, returning to the case of χ having conductor p for simplicity. They consider c_χ as an element of a Galois cohomology group

$$H^1(\mathbb{Q}_p, \mathbb{Q}_p(1)(\overline{\chi}))$$

The Galois module structure of $\mathbb{Q}_p(1)(\overline{\chi})$ is given by $\sigma x = \chi_{\mathrm{cycl}}(\sigma)\overline{\chi}(\sigma)x$, where χ_{cycl} is the standard cyclotomic character. Let $K = \mathbb{Q}_p(\chi)$, then the inflation-restriction on $K/\mathbb{Q}_p$ gives the exact sequence

$$0 = H^1(K/\mathbb{Q}_p, \mathbb{Q}_p(1)(\overline{\chi})^{G_K}) \longrightarrow H^1(\mathbb{Q}_p, \mathbb{Q}_p(1)(\overline{\chi})) \longrightarrow H^1(K, \mathbb{Q}_p(1)(\overline{\chi}))^{\mathrm{Gal}(K/\mathbb{Q}_p)} \longrightarrow$$

$$\longrightarrow H^1(K/\mathbb{Q}_p, \mathbb{Q}_p(1)(\overline{\chi})^{G_K}) = 0.$$

Therefore $H^1(\mathbb{Q}_p, \mathbb{Q}_p(1)(\overline{\chi})) \cong H^1(K, \mathbb{Q}_p(1)(\overline{\chi}))^{\mathrm{Gal}(K/\mathbb{Q}_p)}$. We can then compute by Kummer theory that

$$H^1(K, \mathbb{Q}_p(1)(\overline{\chi}))^{\mathrm{Gal}(K/\mathbb{Q}_p)} \cong H^1(K, \mathbb{Q}_p(1))(\overline{\chi})^{\mathrm{Gal}(K/\mathbb{Q}_p)} \cong \mathbb{Q}_p \otimes \check{K}^\times(\overline{\chi})^{\mathrm{Gal}(K/\mathbb{Q}_p)} \cong \varepsilon_\chi \check{K}^\times.$$

So their Galois cohomology groups are just rational versions of our multiplicative groups. If one used $\mathbb{Z}_p(1)(\overline{\chi})$ coefficients instead of $\mathbb{Q}_p(1)(\overline{\chi})$ coefficients, then the same argument would go through and one would obtain the χ-part of multiplicative group.

They then view the logarithm as mapping into the Dieudonné module (with period ring $\mathbb{C}_p$) of $\mathbb{Q}_p(1)$. They intepret the sum of $\log_p$ in $L_p(\chi, 1)$ as coming from $\log(c_\chi)$, and they interpret the Gauss sum as a period of this vector space.

Now, where we differ from them is that we show that the Gauss sum in the formula for $L_p(\chi, 1)$ can *also* come from the logarithm of a unit, that being u_χ. So our results are essentially an integral version of their results which gives further elaboration on where the Gauss sum "periods" come from.

They further go on to obtain analogues of their result using $\mathbb{Q}_p(n)$ for varying n, and Galois representations associated with elliptic curves or modular forms. It would be interesting to see if our integral explanation of the periods can also be extended to these more general Galois representations.

Along the way to obtaining these main results, we study a curious property of the image of ϱ. It gives what are called "Volkenborn" distributions, which are distributions μ on $\mathbb{Z}_p$ that satisfy a bound of the form

$$|\mu(a + p^n\mathbb{Z}_p) - p\mu(a + p^{n+1}\mathbb{Z}_p)|_p \leq C$$

for some fixed constant C. These are interesting because one can define an integration theory for differentiable functions against Volkenborn integrals. Because of this, we take the opportunity in Section 2 and Appendix A to go through some of the main theorems on Volkenborn integration. We give new proofs of convergence that take advantage of a p-adic Taylor's theorem instead of using Mahler series like the usual proof.

However, the author believes that the fact that ϱ produces Volkenborn distributions is largely unrelated to the story of the construction of u_χ, but there does seem to be more potential here. For example, ϱ produces Volkenborn distributions even when you do not work with the cyclotomic $\mathbb{Z}_p$ extension, so there are a lot of cases left unexplored.

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2 Volkenborn integration and a formula for p-adic L-functions

2.1 p-adic measures

Let $C^0(\mathbb{Z}_p, \mathbb{C}_p)$ (or $C^0(\mathbb{Z}_p)$) denote the vector space of continuous functions from the p-adic integers to the p-adic complex numbers. A classic question one may ask about $C^0(\mathbb{Z}_p)$ is an explicit description of its dual space, and the answer to this is well known and is given by measures. We define a distribution on $\mathbb{Z}_p$ to be a finitely additive function from the set of clopen subsets of $\mathbb{Z}_p$ to $\mathbb{C}_p$. Such a function can be uniquely described by its restriction to sets of the form $a + p^n\mathbb{Z}_p$, where $a \in \mathbb{Z}_p$ and $n \in \mathbb{Z}_{\geq 0}$. More generally, we may think of a distribution as taking values in a ring R, for example $\mathbb{Z}_p$ or a local field K. We describe this space as $\mathcal{D}(\mathbb{Z}_p, R)$.

Given a distribution, we say that it is a measure if it is bounded: that is, $|\mu(a + p^n\mathbb{Z}_p)|_p \leq C$ for some fixed constant C. The space of $\mathbb{C}_p$-valued measures is written as $\mathcal{M}(\mathbb{Z}_p)$ and if they take values in a ring R (where R is required to have some notion of p-adic absolute value) then we write $\mathcal{M}(\mathbb{Z}_p, R)$.

Definition 2.1. *The set of finitely additive functions to a ring R on the clopen subsets of $\mathbb{Z}_p$ is denoted $\mathcal{D}(\mathbb{Z}_p, R)$. The space of such additive functions that are bounded is denoted $\mathcal{M}(\mathbb{Z}_p, R)$. If R is omitted, it is assumed to be $\mathbb{C}_p$.*

There are many different ways to understand $\mathcal{M}(\mathbb{Z}_p, R)$, one perspective comes from the Amice or Mahler transform, which will relate it to an R-valued power series.

Definition 2.2. *Let R be a complete subring of $\mathbb{C}_p$ containing $\mathbb{Z}_p$. Then for $\mu \in \mathcal{M}(\mathbb{Z}_p, R)$, define the Amice transform*

$$f_\mu(T) = \int_{\mathbb{Z}_p} (1+T)^x d\mu(x)$$

where the integration is p-adic Lebesgue integration.

We have the following well-known results in p-adic analysis

Proposition 2.3. *Let R be a subring of $\mathbb{C}_p$ containing $\mathbb{Z}_p$. The Amice transform induces an isomorphism*

$$\mathcal{M}(\mathbb{Z}_p, R) \cong R \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\![T]\!].$$

Proof. See [RJW25, Def. 3.8] □

Proposition 2.4. *The dual space of $C^0(\mathbb{Z}_p)$ is $\mathcal{M}(\mathbb{Z}_p)$.*

Proof. See [RJW25, Rem. 3.12] □

2.2 Volkenborn distributions

We want to find a generalization of this theory that describes the dual space of differentiable functions. Given $f \in C^0(\mathbb{Z}_p, \mathbb{C}_p)$, we may define a derivative

$$\frac{df}{dx}(z) = \lim_{x,y\to z} \frac{f(x) - f(y)}{x - y}$$

whenever such an expression is convergent. We note that a definition of this kind is typically called a "strong derivative" over the real numbers, but it is the standard definition of the derivative over the p-adics. We denote by $S^1(\mathbb{Z}_p, \mathbb{C}_p)$ (or $S^1(\mathbb{Z}_p)$) the subspace of $C^0(\mathbb{Z}_p)$ of differentiable functions. (We note that being differentiable with respect to the strong derivative implies continuous differentiability [Rob00, Sec. 5 Prop. 2].)

We want to give $C^0(\mathbb{Z}_p)$ and $S^1(\mathbb{Z}_p)$ the structure of a Banach space, as it will be convenient to have the standard theorems of functional analysis at our disposal. Defining a norm on $C^0(\mathbb{Z}_p)$ is quite simple, we set

$$\|f\|_0 = \sup_{x\in\mathbb{Z}_p} |f(x)|_p = \max_{x\in\mathbb{Z}_p} |f(x)|_p$$

where $|\cdot|_p$ denotes the p-adic absolute value. The second equality follows from the fact that $\mathbb{Z}_p$ is compact. Defining a Banach space structure on $S^1(\mathbb{Z}_p)$ is a more subtle problem. The usual real-valued approach is to take the C^0 norm of f and its derivative, and then sum them. This does not work p-adically, essentially due to the lack of mean value theorem for differentiable functions.

Instead, we consider the difference quotient operator $\Phi : S^1(\mathbb{Z}_p) \to C^0(\mathbb{Z}_p \times \mathbb{Z}_p)$ given by

$$(\Phi f)(x, y) = \begin{cases} \frac{f(x)-f(y)}{x-y} & \text{if } x \neq y; \\ \frac{df}{dx}(x) & \text{if } x = y. \end{cases}$$

We define the norm on $C^0(\mathbb{Z}_p \times \mathbb{Z}_p)$ the same way we defined the norm on $C^0(\mathbb{Z}_p)$, and then we define the S^1-norm $\|\cdot\|_1$ by

$$\|f\|_1 = \max\{f(0), \|\Phi f\|_0\}.$$

The choice of 0 in $f(0)$ is arbitrary, and indeed one can show that no matter what point we evaluate at, we get an identical norm. See [Rob00, Sec. 5.1] for further discussion. We record the proceeding discussion in the following

Definition 2.5. *The space of strictly differentiable functions, denoted $S^1(\mathbb{Z}_p)$, is the infinite dimensional $\mathbb{C}_p$-Banach space of continuous functions with a well-defined strict derivative. Its norm is*

$$\|f\|_1 = \max\{f(0), \|\Phi f\|_0\}$$

where $\Phi : S^1(\mathbb{Z}_p) \to C^0(\mathbb{Z}_p \times \mathbb{Z}_p)$ is the difference quotient operator.

We want to define certain functionals on $S^1(\mathbb{Z}_p)$. First, say we are given a $\mathbb{C}_p$-valued measure μ on $\mathbb{Z}_p$. This allows us to define a functional on $C^0(\mathbb{Z}_p)$ via Lebesgue integration

$$f \mapsto \int_{\mathbb{Z}_p} f(x)d\mu(x).$$

These functionals all restrict to functionals on $S^1(\mathbb{Z}_p)$, however these are not suitable for our arithmetic applications. Instead, we consider a different integration theory, built around a larger set of distributions.

Definition 2.6. *Let μ be a (usually unbounded) $\mathbb{C}_p$-valued distribution on $\mathbb{Z}_p$. We say that μ is **Volkenborn** if there is some absolute constant $C_\mu \in \mathbb{R}_{>0}$ such that*

$$|\mu(a + p^n\mathbb{Z}_p) - p\mu(a + p^{n+1}\mathbb{Z}_p)|_p \leq C_\mu.$$

We let $\mathcal{V}(\mathbb{Z}_p)$ denote the space of Volkenborn distributions on $\mathbb{Z}_p$.

Remarks.

1. Every p-adic measure on $\mathbb{Z}_p$ is also a Volkenborn distribution.

2. The Haar distribution $\mathbf{h}(a + p^n\mathbb{Z}_p) = 1/p^n$ is a Volkenborn distribution. In fact, one may choose $C_{\mathbf{h}} = 0$, so in some sense, Volkenborn distributions are "approximately Haar."

3. The space of $\mathbb{C}_p$-valued measures, $\mathcal{M}(\mathbb{Z}_p)$, acts on $\mathcal{V}(\mathbb{Z}_p)$ via convolution. For $\mu \in \mathcal{V}(\mathbb{Z}_p)$ and $\lambda \in \mathcal{M}(\mathbb{Z}_p)$, we may define
$$\mu * \lambda(a + p^n\mathbb{Z}_p) = \sum_{\substack{i,j \in \mathbb{Z}/p^n\mathbb{Z} \\ i+j=a}} \mu(i + p^n\mathbb{Z}_p)\lambda(j + p^n\mathbb{Z}_p).$$
One can show that $\mu * \lambda \in \mathcal{V}(\mathbb{Z}_p)$ [AW17, Lem. 3.1], so $\mathcal{V}(\mathbb{Z}_p)$ is a $\mathcal{M}(\mathbb{Z}_p)$-module.

4. Sitting inside $\mathcal{M}(\mathbb{Z}_p)$ we have the ring of $\mathbb{Z}_p$-valued measures on $\mathbb{Z}_p$. This ring is isomorphic to $\Lambda = \mathbb{Z}_p[\![T]\!]$ (see for example [Was12, Ch. 12] or [RJW25, 3.25]), so $\mathcal{V}(\mathbb{Z}_p)$ is in fact a Λ-module as well.

We will describe an integration theory, which we call Volkenborn integration, using Volkenborn distributions. This will generalize the classic theory of Volkenborn integration from [Vol72, Vol74]. Our generalization was first considered by [Bar02].

Definition 2.7. *Let μ be a distribution and $f : \mathbb{Z}_p \to \mathbb{C}_p$. We define the Volkenborn integral of f to be*

$$\int_{\mathbb{Z}_p} f(x)d\mu(x) = \lim_{n\to\infty} \sum_{a=0}^{p^n-1} f(a)\mu(a + p^n\mathbb{Z}_p)$$

if it converges.

This can be viewed as a generalization of the Riemann integral. The Volkenborn has many arithmetic applications that classical p-adic integration does not. For example, if $\mathbf{h}$ is a Haar measure, then one can show that if $\langle \cdot \rangle$ is projection onto the 1-units of $\mathbb{Z}_p^\times$ and $\mathbf{1}_{\mathbb{Z}_p^\times}$ is the indicator function of $\mathbb{Z}_p^\times$ then

$$\zeta_p(s) = \frac{1}{1-s} \int_{\mathbb{Z}_p^\times} \mathbf{1}_{\mathbb{Z}_p^\times}(x) \langle x \rangle^{1-s} d\mathbf{h}(x).$$

The main result in Volkenborn integration is the following

Proposition 2.8. *Let $\mu \in \mathcal{V}(\mathbb{Z}_p)$ and let C_μ be its associated constant. Then Volkenborn integration*

$$I_\mu : f \mapsto \int_{\mathbb{Z}_p} f(x) d\mu(x)$$

defines a bounded linear functional from $S^1(\mathbb{Z}_p)$ to $\mathbb{C}_p$. Moreover, if C_μ is the constant associated with μ by Definition 2.6, we can bound the operator norm as $\|I_\mu\|_{op} \leq p \max\{C_\mu, |\mu(\mathbb{Z}_p)|_p\}$.

Proof. See Section A, specifically Lemmas A.4 and A.5. For the original proof using Mahler series, see [Bar02, Thm. 2.4.1]. ◻

We will use Volkenborn integration to define an Amice transform on all of $\mathcal{D}(\mathbb{Z}_p)$. Let μ be an arbitrary distribution and define a function of z, for all values of z where it converges

$$f_\mu(z) = \int_{\mathbb{Z}_p} (1+z)^x d\mu(x).$$

We are interested in certain evaluations of $f_\mu(z)$ that we will call its "special values" $f_\mu(\zeta_{p^m}^b - 1)$. We will see later that in some sense that the map $\mu \mapsto f_\mu(z)$ is in some sense better behaved when one restricts it to the special values, as these are the values where we can always guarantee convergence.

Definition 2.9. *The set of $\mathbb{C}_p$-valued functions defined on the set $\{\zeta_{p^m}^b - 1 : m, b \in \mathbb{N}\}$ is denoted $\mathcal{F}(\mu_{p^\infty} - 1)$. Both of these are $\mathbb{C}_p$-algebras with multiplication on the first is given by convolution, and multiplication on the second is defined pointwise.*

Lemma 2.10. *Let $\mu \in \mathcal{D}(\mathbb{Z}_p)$. Then*

$$f_\mu(\zeta_{p^m}^b - 1) = \sum_{a=0}^{p^m-1} \zeta_{p^m}^{ab} \mu(a + p^m\mathbb{Z}_p).$$

Proof. The key idea is that the limit in Definition 2.7 stabilizes for $n \geq m$. Observe that for $n \geq m$ we have the following computation of $f_\mu(\zeta_{p^n}^b - 1)$:

$$\begin{aligned}
\sum_{a=0}^{p^n-1} (\zeta_{p^m}^b - 1 + 1)^a \mu(a + p^m\mathbb{Z}_p) &= \sum_{a=0}^{p^n-1} \zeta_{p^m}^{ab} \mu(a + p^m\mathbb{Z}_p) \\
&= \sum_{a_1=0}^{p^m-1} \sum_{a_2=0}^{p^{n-m}-1} \zeta_{p^m}^{(a_1+p^m a_2)b} \mu(a_1 + p^m a_2 + p^n\mathbb{Z}_p) \\
&= \sum_{a_1=0}^{p^m-1} \zeta_{p^m}^{a_1 b} \mu(a_1 + p^m\mathbb{Z}_p).
\end{aligned}$$

◻

Remark. The specific choice of evaluating f_μ at $\zeta_{p^n}^b - 1$ is notable as these are the only choices of T that make $(T+1)^x$ a locally constant function. This ensures convergence of the limit for any distribution.

For the moment, we bring back the coefficient ring as we will need it in sections 5 and 6

Lemma 2.11. *Let K be a $\mathbb{Q}_p(\zeta_{p^\infty})$ algebra. The Amice transform $\mu \mapsto f_\mu$ is a ring isomorphism between $\mathcal{D}(\mathbb{Z}_p, K)$ and $\mathcal{F}(\mu_{p^\infty} - 1, K)$.*

Proof. The following proof is similar to proofs that the discrete Fourier transform on $\mathbb{Z}/p^n\mathbb{Z}$ is an isomorphism from the ring of such functions under convolution to the ring of such functions under pointwise multiplication.

The fact that the Amice transform is a ring homomorphism is clear and follows from direct calculation. We construct its inverse. For $f \in \mathcal{F}(\mu_{p^\infty} - 1, K)$, define $\mu \in \mathcal{D}(\mathbb{Z}_p, K)$ by the formula

$$\mu(a + p^n\mathbb{Z}_p) = \frac{1}{p^n}\sum_{b=0}^{p^n-1} \zeta_{p^n}^{-ab} f(\zeta_{p^n}^b - 1).$$

We verify (using Lemma 2.10 in the first equality) that

$$\begin{aligned} f_\mu(\zeta_{p^n}^a - 1) &= \sum_{b=0}^{p^n-1} \zeta_{p^n}^{ab}\mu(b + p^n\mathbb{Z}_p) \\ &= \frac{1}{p^n}\sum_{b=0}^{p^n-1} \zeta_{p^n}^{ab} \sum_{c=0}^{p^n-1} \zeta_{p^n}^{-bc} f(\zeta_{p^n}^c - 1) \\ &= \frac{1}{p^n}\sum_{b=0}^{p^n-1}\sum_{c=0}^{p^n-1} \zeta_{p^n}^{ab-bc} f(\zeta_{p^n}^c - 1) = f(\zeta_{p^n}^a - 1). \end{aligned}$$

□

Now we are interested in what happens when we restrict μ to $\mathcal{V}(\mathbb{Z}_p)$. Working formally, we see that f_μ "should" be a power series in z. Replacing z with T to indicate that this only holds formally, we have by the generalized binomial theorem

$$f_\mu(T) = \int_{\mathbb{Z}_p} \sum_{r=0}^{\infty} \binom{x}{r} T^r d\mu(x) = \sum_{r=0}^{\infty} \left(\int_{\mathbb{Z}_p} \binom{x}{r} d\mu(x) \right) T^r.$$

This will turn out to actually be true when we evaluate the power series.

Proposition 2.12. *Let μ be a Volkenborn distribution, then for any $z \in \mathbb{C}_p$ with $|z|_p < 1$, $f_\mu(z)$ converges. We also have*

$$f_\mu(z) = \sum_{r=0}^{\infty} z^r \int_{\mathbb{Z}_p} \binom{x}{r} d\mu(x).$$

Proof. See Section A or [Bar02, Sec. 2.3] □

Going forward, we will assume $\mu \in \mathcal{V}(\mathbb{Z}_p)$. We want to compare the formal power series $f_\mu(T)$ and the element $f_\mu(z) \in \mathcal{F}(\mu_{p^\infty} - 1)$. To do this we establish some notation

Definition 2.13. *The $\mathbb{C}_p$-algebra of power series convergent in $\{z \in \mathbb{C}_p : |z| < 1\}$ is denoted by $H(\mathbb{C}_p)$.*

We have a commutative diagram of $\mathbb{C}_p$-vector spaces

$$\begin{array}{ccc} \mathcal{V}(\mathbb{Z}_p) & \xrightarrow{\mu \mapsto f_\mu(T)} & H(\mathbb{C}_p) \\ & {\scriptstyle \mu \mapsto f_\mu(z)} \searrow & \downarrow {\scriptstyle \mathrm{ev}_{\mu_{p^\infty} - 1}} \\ & & \mathcal{F}(\mu_{p^\infty} - 1). \end{array}$$

We note that each of these $\mathbb{C}_p$-vector spaces have the structure of $\mathcal{M}(\mathbb{Z}_p)$-modules. The action on $\mathcal{V}(\mathbb{Z}_p)$ is via convolution. There is an isomorphism $\mathcal{M}(\mathbb{Z}_p) \cong \mathbb{C}_p \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\![T]\!]$ by Proposition 2.3, so we also have a $\mathcal{M}(\mathbb{Z}_p)$ action on $H(\mathbb{C}_p)$ and $\mathcal{F}(\mu_{p^\infty} - 1)$ by pointwise multiplication.

It is a very important question whether or not the maps in this diagram are homomorphisms of $\mathcal{M}(\mathbb{Z}_p)$-modules. By the "ring homomorphism" part of Lemma 2.11, we know that the diagonal map $\mu \mapsto f_\mu(z)$ is a module homomorphism. On the other hand, the vertical map $\mathrm{ev}_{\mu_{p^\infty}-1}$ is also a module homomorphism, since the $\mathcal{M}(\mathbb{Z}_p)$-action on both spaces are the same. However, it is not obvious that $\mu \mapsto f_\mu(T)$ is a ring homomorphism, due to the extreme delicateness of the convergence in Volkenborn integration. In fact, it turns out to *not* be a $\mathcal{M}(\mathbb{Z}_p)$-module homomorphism. Soon we will state a theorem that measures the extent to which it fails to be a $\mathcal{M}(\mathbb{Z}_p)$-module homomorphism, but before then we describe a simple counterexample.

Let $\mathbf{h}$ be the Haar distribution, and $\lambda = \delta_0 - \delta_1$ be a difference of delta measures. A simple calculation confirms that $\mathbf{h} * \lambda = 0$. If $\mu \mapsto f_\mu(T)$ preserved multiplication, then this would imply that $f_{\mathbf{h}}(T) f_\lambda(T) = 0$. We know that $H(\mathbb{C}_p)$ is a domain, so since the $\mathcal{M}(\mathbb{Z}_p)$-action comes from the subring $\mathbb{C}_p \otimes_{\mathcal{O}_{\mathbb{C}_p}} \mathcal{O}_{\mathbb{C}_p}[\![T]\!] \subseteq H(\mathbb{C}_p)$, $f_{\mathbf{h}} f_\lambda = 0$ implies $f_\mu(T) = 0$ or $f_\lambda(T) = 0$. A straightforward calculation gives $f_\lambda(T) = T$, so we just need to see if $f_{\mathbf{h}}(T) = 0$.

Proposition 2.14. *The Amice transform of* $\mathbf{h}$ *is given by*

$$f_{\mathbf{h}}(T) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r+1} T^r = \frac{\log(1+T)}{T}$$

Proof. We need to compute the coefficients of $f_{\mathbf{h}}$, which by Proposition 2.12 is given by integrating binomial coefficients against $\mathbf{h}$. We see that

$$\begin{aligned}
\int_{\mathbb{Z}_p} \binom{x}{r} d\mathbf{h}(x) &= \lim_{n\to\infty} \sum_{a=0}^{p^n-1} \binom{a}{r} \frac{1}{p^n} \\
&= \lim_{n\to\infty} \frac{1}{p^n} \binom{p^n}{r+1} \\
&= \lim_{n\to\infty} \frac{p^n(p^n-1)\cdots(p^n-r)}{p^n(r+1)!} \\
&= \lim_{n\to\infty} \frac{(-1)^r}{r+1} + O(p^n) = \frac{(-1)^r}{r+1}.
\end{aligned}$$

The result follows. ∎

Since both $f_{\mathbf{h}}(T)$ and $f_\lambda(T)$ are not zero, it follows that $\mu \mapsto f_\mu(T)$ is not a $\mathcal{M}(\mathbb{Z}_p)$-module homomorphism.

We can actually describe precisely the difference between $f_{\mu*\lambda}(T)$ and $f_\mu(T) f_\lambda(T)$. To do this, we must introduce some notation. For a Volkenborn distribution μ, define a function $F_\mu : \mathbb{Z}_p \to \mathbb{C}_p$ by

Definition 2.15. *For a Volkenborn distribution* $\mu \in \mathcal{V}(\mathbb{Z}_p)$*, define its Radon-Nikodym derivative*

$$F_\mu(x) = \lim_{n\to\infty} p^n \mu(x + p^n \mathbb{Z}_p).$$

This is a continuous function and is even 0 *if* μ *is a measure[Bar02, Def. 3.1].*

In the following, let $\mathbf{S} : C^0(\mathbb{Z}_p) \to C^0(\mathbb{Z}_p)$ denote the inverse of the forward difference operator. Specifically, for $x \in \mathbb{Z}_{\geq 0}$ we may define $(\mathbf{S}f)(x) = \sum_{0 \leq a < x} f(x)$ and define $f(x)$ for $x \in \mathbb{Z}_p$ by continuity. (This does yield an element of $C^0(\mathbb{Z}_p)$, see [Rob00, Sec. 4.2.5, Cor. 2].)

Proposition 2.16. *Let* $\mu \in \mathcal{V}(\mathbb{Z}_p)$ *and* $\lambda \in \mathcal{M}(\mathbb{Z}_p)$*. Then*

$$f_{\mu*\lambda}(T) = f_\mu(T) f_\lambda(T) - \log_p(1+T) \sum_{r=0}^{\infty} \int_0^\infty \mathbf{S}^{r+1}(F_\mu(-1-x)) d\lambda(x) T^r.$$

Proof. See [AW17, Prop. 3.2] ∎

Corollary 2.17. $f \mapsto f_\mu(T)$ *does not define a homomorphism of* $\mathcal{M}(\mathbb{Z}_p)$*-modules from* $\mathcal{V}(\mathbb{Z}_p)$ *to* $H(\mathbb{C}_p)$*, but it does define one from* $\mathcal{V}(\mathbb{Z}_p)$ *to* $H(\mathbb{C}_p)/(\log_p(1+T))$

One can view Proposition 2.17 as a refinement of the fact that $\mu \mapsto f_\mu(z)$ is a homomorphism of $\mathcal{M}(\mathbb{Z}_p)$-modules $\mathcal{V}(\mathbb{Z}_p) \to \mathcal{F}(\mu_{p^\infty} - 1)$, since the kernel of the evaluation map $\mathrm{ev}_{\mu_{p^\infty}-1} : H(\mathbb{C}_p) \to \mathcal{F}(\mu_{p^\infty} - 1)$ clearly contains $\log_p(1+T)$. However, it is not clear whether this is a true refinement in the following sense: we do not know if $(\log_p(1+T)) = \ker(\mathrm{ev}_{\mu_{p^\infty}} - 1)$. The author is not confident one way or the other, and so we pose the following

Question. Does $(\log_p(1+T)) = \ker(\mathrm{ev}_{\mu_{p^\infty}} - 1)$?

There are a couple things that make this difficult. First, the fact that we are working in $H(\mathbb{C}_p)$ instead of $\mathbb{C}_p\langle T\rangle$ or $\mathbb{C}_p \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\![T]\!]$ means we do not have access to any form of Weierstrass preparation. To make matters worse, $\{x \in \mathbb{C}_p : |x| < 1\}$, while closed, is not compact. This makes topological arguments quite difficult.

2.3 Special values of p-adic L-functions

In this section, we will do some computations involving p-adic L-functions that will be necessary for the main results of this paper, specifically Theorem 5.6 and Theorem 6.6.

Let $m = pm'$ be a positive integer divisible by p exactly once. Let χ be a character of $(\mathbb{Z}/m\mathbb{Z})^\times$ with conductor m. Let ψ be a purely wild character at p, i.e. a character of $(\mathbb{Z}/p^{n+1}\mathbb{Z})^\times$ that is trivial on $(\mathbb{Z}/p\mathbb{Z})^\times$. Define

$$s(\chi\psi) = \sum_{a=0}^{mp^n-1} \overline{\chi}(a) \log_p(1 - \zeta_{mp^n}^a).$$

We will need to fix an isomorphism $(1 + p\mathbb{Z}_p, \times) \cong \mathbb{Z}_p$ moving forward. This amounts to choosing a generator, for which $1 + m$ will do (recall the m is divisible by p exactly once.) This choice will actually be quite important for comparing different formulas for p-adic L-functions.

Given a purely wild character ψ, we can uniquely associate a p^nth root of unity ζ_ψ by $\psi(1+m) = \zeta_\psi$. We want to construct a distribution $\mu_{s,\chi}$ such that $f_{\mu_s}(\zeta_\psi - 1) = s(\chi\psi)$.

Definition 2.18. *Let χ and ψ be as above. Define*

$$\mu_{s,\chi}(a + p^n\mathbb{Z}_p) = \sum_{b \in \mathbb{Z}/m\mathbb{Z}} \chi(b) \log_p(1 - \zeta_{mp^n}^{\omega(b)(1+m)^a})$$

We denote its Fourier transform $f_{\mu_{s,\chi}}$ as $f_{s,\chi}$. The function $\omega : (\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{Z}_p \times (\mathbb{Z}/m'\mathbb{Z})$ works by decomposing $(\mathbb{Z}/m\mathbb{Z})^\times \cong (\mathbb{Z}/p\mathbb{Z})^\times \times (\mathbb{Z}/m'\mathbb{Z})$, applying the Teichmuller character to the first coordinate, and the identity to the second coordinate.

One can compute that $f_{s,\chi}(\zeta_\psi - 1) = s(\chi\psi)$. Similar to $s(\chi)$ we have the Gauss sums

$$\tau(\chi\psi) = \sum_{a=0}^{mp^n} \chi\psi(a)\zeta_{mp^n}^a.$$

We define the following distribution in terms of the Gauss sums. This construction is done by tracking through the proof of 2.11 to construct a distribution whose amice transform gives Gauss sums.

Definition 2.19. *Let χ and ψ be as above. For a p^nth root of unity $\zeta_{p^n}^a$, define $\psi_{\zeta_{p^n}^a}$ to be the purely wild character with $\psi_{\zeta_{p^n}^a}(1+m) = \zeta_{p^n}^a$. Write*

$$\mu_{\tau,\chi}(a + p^n\mathbb{Z}_p) = \frac{1}{p^n} \sum_{b=0}^{p^n-1} \zeta_{p^n}^{-ab} \tau(\overline{\zeta\psi_{\zeta_{p^n}^a}})$$

We denote its Fourier transform $f_{\mu_{\tau,\chi}}$ as $f_{\tau,\chi}$.

A corollary of the proof of Lemma 2.11 gives $f_{\tau,\chi}(1-\zeta_\psi) = \tau(\chi\psi)$. We observe the following equality as a result of the special value formula for p-adic L-functions at 1, see [Was12, Thm. 5.18].

$$f_{s,\overline{\chi}}(1-\overline{\zeta_\psi}) = -L_p(\chi\psi,1)f_{\tau,\overline{\chi}}(1-\overline{\zeta_\psi}).$$

We want to relate the quotient $f_{s,\chi}(T)/f_{t,\chi}(T)$ to the p-adic power series that correspond to $L_p(\chi,s)$. To do this, we need the following two power series

Definition 2.20. *Let χ be as before. Define $F_\chi \in \mathbb{Z}_p[\chi(1),\cdots,\chi(m-1)]\otimes\Lambda$ such that*

$$F_\chi(\zeta_\psi(1+m)^s-1) = L_p(\chi\psi,s).$$

This is known to exist by [Was12, Thm. 7.10].

We can then define $G_\chi(T) = F_\chi(\frac{1+m}{1+T}-1)$, and can compute $L_p(\chi\psi,1) = G_\chi(\overline{\zeta_\psi}-1)$, so we ultimately obtain

$$f_{\overline{\chi},s}(\overline{\zeta_\psi}-1) = -G_\chi(\overline{\zeta_\psi}-1)f_{\overline{\chi},t}(\overline{\zeta_\psi}-1). \tag{3}$$

Theorem 2.21. *Let $\lambda_{\chi,p}\in\mathcal{M}(\mathbb{Z}_p)$ be such that $f_{\lambda_p}(T) = G_\chi(T)$. Then we have an equality of distributions*

$$\mu_{s,\chi} = -\lambda_{p,\overline{\chi}} * \mu_{\tau,\chi}.$$

Proof. We recall the isomorphism $\mathcal{D}(\mathbb{Z}_p)\cong\mathcal{F}(\mu_{p^\infty}-1)$ from Lemma 2.11, so this follows from Equation 3. $\square$

To prove the second case of Theorem 5.6, we will need another result. First define $\dot{T} = \frac{1+m}{1+T}-1$. Let $\dot{\mu}$ be the measure associated to $\dot{T}$ by the Amice transform (see Proposition 2.3).

Definition 2.22. *Define $\mu_{r,\chi} = \dot{\mu}*\mu_{\tau,\chi}$. We can alternatively define it from its Amice transform by*

$$f_{\mu_r}(\zeta_\psi-1) = \overline{\zeta_\psi}(1+m)\tau(\chi\psi).$$

Corollary 2.23. *Let $\tilde{\lambda}_{p,\chi}$ be such that $f_{\tilde{\lambda}_{p,\chi}}(T) = G_\chi(T)/\dot{T}$. Then*

$$\mu_{s,\chi} = -\tilde{\lambda}_{p,\overline{\chi}} * \mu_{r,\chi}.$$

3 Lemmas from representation theory

In sections 4, 5, and 6, we will be faced with the systematic study of modules over $\mathbb{Z}_p[G]$ where G is a finite abelian group. To enable us, it will useful to have a toolkit of "$\mathbb{Z}_p$-valued" representation theory. We will see that the case where $p \mid |G|$ and $p\nmid |G|$ are fundamentally quite different, essentially for the same reason that representation theory over fields is difficult when the characteristic divides the order of the group.

Another issue that will challenge us is the fact that $\mathbb{Z}_p$ is not integrally closed. This will mean that there will be representations that are decomposable upon tensoring with $\mathcal{O}_{\overline{\mathbb{Q}_p}}$, but are irreducible in $\mathbb{Z}_p$.

3.1 The χ-part and χ-quotient

Let χ be a character of G taking values in $\overline{\mathbb{Q}_p}$. There is a certain representation canonically associated to χ. To describe it, we first let $\mathbb{Z}_p[\chi] := \mathbb{Z}_p[\chi(g) : g\in G]$. It will be important in the future to make sure we do not think of $\mathbb{Z}_p[\chi]$ as a valuation ring of a local field. It should only be considered as a coefficient ring.

Definition 3.1. *Let $\mathbb{Z}_p(\chi)$ denote the $\mathbb{Z}_p[G]$-module with underlying abelian group $\mathbb{Z}_p[\chi]$ and G-action $gx = \chi(g)x$.*

Now, let M be a $\mathbb{Z}_p[G]$-module, $H\subseteq G$, and χ a character of H. We want to describe two types of χ-component of M.

Definition 3.2. *Let M, G, H, χ be as above. We define the χ-part and χ-quotient of M as*

$$M^\chi = Hom_{\mathbb{Z}_p[H]}(\mathbb{Z}_p(\chi), M) \qquad \text{and} \qquad M_\chi = \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p[H]} M.$$

Remark. The χ-part and χ-quotient of M satisfy universal properties. Given a $\mathbb{Z}_p[\chi][G]$-module N, we say that H acts via χ in $hx = \chi(h)x$ for $x \in N$ and $h \in H$. Then for all N where H acts through χ, we can make diagrams of the following form commute

$$\begin{tikzcd} \mathrm{Hom}_{\mathbb{Z}_p}(\mathbb{Z}_p[\chi], M) & M^\chi \arrow[l] \\ & N \arrow[ul] \arrow[u, dashed] \end{tikzcd} \qquad \begin{tikzcd} M_\chi \arrow[d, dashed] & M \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\chi] \arrow[l] \arrow[dl] \\ N & \end{tikzcd}$$

We note that since $\mathbb{Z}_p[\chi]$ is free as a $\mathbb{Z}_p$-module, $\mathrm{Hom}_{\mathbb{Z}_p}(\mathbb{Z}_p[\chi], M)$ and $M \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\chi]$ are (noncanonically) isomorphic.

There are three useful lemmas about χ-parts and χ-quotients that we may immediately state.

Lemma 3.3. *The functor $M \to M^\chi$ is left exact and the functor $M \to M_\chi$ is right exact.*

Proof. This follows from the fact that Hom is left exact and $\otimes$ is right exact. ☐

Lemma 3.4. *There are natural isomorphisms of $\mathbb{Z}_p[\chi]$-modules*

$$M^\chi \cong \{m \in M \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\chi] : gx = \chi(g)x\}$$
$$M_\chi \cong (M \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\chi])/I_\chi(M \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\chi])$$

where $I_\chi = ([g] - \chi(g) | g \in G) \subseteq \mathbb{Z}_p[\chi][G]$

Proof. This material can likely be found in many places, but [Tsu99, Lem. 2.1] discusses it in our context. ☐

Lemma 3.5. *Let $G = H_1 \times H_2$, let χ_1, χ_2 be characters of H_1, H_2 and $\chi = \chi_1 \times \chi_2$. Then*

$$(M^{\chi_1})^{\chi_2} = M^\chi \qquad \text{and} \qquad (M_{\chi_1})_{\chi_2} = M_\chi$$

Proof. See [Tsu99]. ☐

Let $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)$. Then in some sense the characters χ and χ^σ should look the same when viewed over $\mathbb{Q}_p$. We can see this philosophy hold true in the following sense

Lemma 3.6. *Let $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)$ and let M be a $\mathbb{Z}_p[G]$-module. Then there are natural isomorphisms of $\mathbb{Z}_p[G]$-modules*

$$M^\chi \cong M^{\chi^\sigma} \qquad \text{and} \qquad M_\chi \cong M_{\chi^\sigma}.$$

Proof. The key point is that $\sigma|_{\mathbb{Z}_p[\chi]}$ induces an isomorphism of $\mathbb{Z}_p[G]$-modules $\mathbb{Z}_p(\chi) \to \mathbb{Z}_p(\chi^\sigma)$. The claimed isomorphisms then follow from functoriality of Hom and $\otimes$. ☐

Remark. With the above result in mind, we see that it makes sense to not just think of M_χ and M^χ as the χ-components of M, but as the χ^σ component for all $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)$. Pushing this idea further, we have an isomorphism

$$\overline{\mathbb{Q}_p} \otimes \mathbb{Z}_p(\chi) \cong \bigoplus_{\psi = \chi^\sigma} \overline{\mathbb{Q}_p}(\psi)$$

where $\overline{\mathbb{Q}_p}(\psi)$ is the irreducible $\overline{\mathbb{Q}_p}$ representation associated with the character ψ. In this sense, we may think of $\mathbb{Z}_p[\chi]$ as being a combination of representations associated to ψ for each ψ conjugate to χ. Alternatively, it can be thought of as the $\sum_{\sigma \in \mathrm{Gal}(\mathbb{Q}_p[\chi]/\mathbb{Q}_p)} \chi^\sigma$-part, and we make this precise when $|\Delta|$ is prime to p in 3.10

Due to the presence of tensor products and homs over a non-semisimple ring, one may suspect that we will end up needing to consider Ext and Tor groups. This is true, and to compute them we will be helped by this lemma.

Lemma 3.7. *Let R be a ring and G a finite group. Let M, N be $R[G]$-modules. There are spectral sequences of Homological and Cohomological type respectively*

$$E^2_{pq} = H_p(G, \mathrm{Tor}^R_q(M,N)) \Longrightarrow \mathrm{Tor}^{R[G]}_{p+q}(M,N)$$
$$E^{pq}_2 = H^p(G, \mathrm{Ext}^q_R(M,N)) \Longrightarrow \mathrm{Ext}^{p+q}_{R[G]}(M,N)$$

Remark. In this statement we take a particular convention for the G-action on $M \otimes N$. Given $g \in G, m \in M$, and $n \in N$, we define $g(m \otimes n) = g^{-1}m \otimes gn$.

Proof. This follows from the Grothendieck spectral sequence, see [NSW08, Thm. 2.6.10]. □

When $p \nmid |G|$ (or more generally when $p \nmid |H|$ for some subgroup of G) the concepts of χ-part and χ-quotient turn out to coincide, becoming much simpler. To see why, set $R = \mathbb{Z}_p$ in the above lemma observe that if $p \nmid |G|$, then $H^p(G, \mathrm{Ext}^q_{\mathbb{Z}_p}(M,N)) = 0$ for $q > 0$ because multiplication by $|G|$ will be invertible on the $\mathbb{Z}_p$-module $\mathrm{Ext}^q_{\mathbb{Z}_p}(M,N)$ (see [NSW08, Thm. 2.6.12].) This means the relevant spectral sequence degenerates on the 2nd page and one obtains canonical isomorphisms $\mathrm{Ext}^p_{R[G]}(M,N) \cong \mathrm{Ext}^p_R(M,N)^G$. A similar thing happens to Tor groups.

Here is the practical affect of this.

Definition 3.8. *Let $H \subseteq G$ with $p \nmid |H|$ and let χ be a character of H and let $\overline{\chi}$ be its inverse. We define the **idempotent** $\varepsilon_\chi \in \mathbb{Z}_p[\chi][G]$*

$$\varepsilon_\chi = \frac{1}{|H|} \sum_{h \in H} \overline{\chi}(h)[h].$$

Proposition 3.9. *Idempotents have the following properties:*

1. *They're actually idempotents: $\varepsilon_\chi^2 = \varepsilon_\chi$,*
2. *If $\chi \neq \psi$ are both characters of H, then $\varepsilon_\chi \varepsilon_\psi = 0$,*
3. *They act like Eigenvectors: for all $h \in H$, $h\varepsilon_\chi = \chi(h)\varepsilon_\chi$,*
4. *$M \cong \varepsilon_\chi M \oplus (1 - \varepsilon_\chi)M$*
5. *They unify χ-parts and χ-quotients: $M^\chi \cong \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes_{\mathbb{Z}_p} M) \cong M_\chi$,*
6. *The functor $M \mapsto \varepsilon_\chi M$ is exact,*
7. *The functor $M \mapsto \varepsilon_\chi M$ preserves injcetives and projectives*
8. *For all i we have canonical isomorphisms*

$$\varepsilon_\chi \mathrm{Ext}^i_{\mathbb{Z}_p[G]}(M,N) \cong \mathrm{Ext}^i_{\mathbb{Z}_p[G]}(\varepsilon_\chi M, N) \cong \mathrm{Ext}^i_{\mathbb{Z}_p[G]}(M, \varepsilon_\chi N)$$
$$\varepsilon_\chi \mathrm{Tor}^{\mathbb{Z}_p[G]}_i(M,N) \cong \mathrm{Tor}^{\mathbb{Z}_p[G]}_i(\varepsilon_\chi M, N) \cong \mathrm{Tor}^{\mathbb{Z}_p[G]}_i(M, \varepsilon_\chi N).$$

Proof. (1), (2), and (3) are simple calculations using orthogonality of characters. (4) is because $m = \varepsilon_\chi m + (1 - \varepsilon_\chi)m$ and because $(1 - \varepsilon_\chi)$ is also an idempotent element. (5) can be done by explicitly writing out the isomorphisms, the key conceptual reason it works is because $\varepsilon_\chi M$ is both a subobject and a quotient of M. (6), (7), and (8) actually hold for any idempotent element of any ring, the key is multiplying by an idempotent ε is the same quotienting by $(1 - \varepsilon)$ which is the same inverting ε (see[Sta25, Lem. 10.21.5]). But localization satisfies properties (6), (7), and (8), so we are done. □

Suppose χ is a character of H valued in some extension $\mathbb{Z}_p[\chi]$. Define $\psi : H \to \mathbb{Z}_p$ by

$$\psi(h) = \sum_{\sigma \in \mathrm{Gal}(\mathbb{Q}_p[\chi]/\mathbb{Q}_p)} \chi(h)^\sigma.$$

We once again obtain an idempotent

$$\varepsilon_\psi = \sum_{h\in H} \overline{\psi}(h)[h] = \sum_{\sigma\in \mathrm{Gal}(\mathbb{Q}_p[\chi]/\mathbb{Q}_p)} \varepsilon_{\chi^\sigma}.$$

One can verify that ε_ψ is an idempotent element by using (1) and (2) of Proposition 3.9.

Lemma 3.10. *Let $H \subseteq G$ with $p \nmid |H|$ and χ a character of H, and let ψ and ε_ψ be as described above. Then the trace map $T : \mathbb{Z}_p[\chi] \to \mathbb{Z}_p$ induces an isomorphism*

$$\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes M) \cong \varepsilon_\psi M$$

Proof. Define $\mathcal{G} = \mathrm{Gal}(\mathbb{Q}_p[\chi]/\mathbb{Q}_p)$ and let $\sigma \in \mathcal{G}$. Then if $m \in \varepsilon_\chi \mathbb{Z}_p[\chi] \otimes M$ we have for $h \in H$

$$h(m^\sigma) = (hm)^\sigma = (\chi(h)m)^\sigma = \chi(h)^\sigma m^\sigma = \chi^\sigma(h)m^\sigma.$$

The key point here is that the action of $\mathcal{G}$ is compatible with the G-action. It follows that $m^\sigma \in \varepsilon_{\chi^\sigma} M$. Now, let $T : \mathbb{Z}_p[\chi] \otimes M \to M$ be induced by the trace map. We have

$$T(m) = \sum_{\sigma\in\mathcal{G}} m^\sigma \in \bigoplus_{\sigma\in\mathcal{G}} \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes M) = \varepsilon_\psi(\mathbb{Z}_p[\chi] \otimes M).$$

It follows that $\mathrm{im}(T) \subseteq \varepsilon_\psi(\mathbb{Z}_p[\chi] \otimes M) \cap M = \varepsilon_\psi M$.

Now we will construct a two-sided inverse to T. Consider the inclusion map

$$\iota : \varepsilon_\psi M \to \varepsilon_\psi(\mathbb{Z}_p[\chi] \otimes M)$$

induced by the inclusion $\mathbb{Z}_p \to \mathbb{Z}_p[\chi]$. Then we have the multiplication by ε_χ map $[\varepsilon_\chi] : \varepsilon_\psi(\mathbb{Z}_p[\chi] \otimes M) \to \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes M)$. We now have a (noncommutative!) diagram of maps

$$\begin{tikzcd} \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes M) \arrow[r, "T"] & \varepsilon_\psi(M) \arrow[d, "\iota"] \\ & \varepsilon_\psi(\mathbb{Z}_p[\chi] \otimes M). \arrow[ul, "{[\varepsilon_\chi]}"] \end{tikzcd}$$

Let $S = [\varepsilon_\chi] \circ \iota$. We want to show that $TS = ST = \mathrm{id}$. Let $m \in \varepsilon_\psi M$. We have

$$TSm = T\varepsilon_\chi(1 \otimes m) = \sum_{\sigma\in\mathcal{G}} \varepsilon_\chi^\sigma (1\otimes m)^\sigma = \sum_{\sigma\in\mathcal{G}} \varepsilon_{\chi^\sigma}(1 \otimes m) = \varepsilon_\psi m = m.$$

On the other hand, for $m \in \varepsilon_\chi \mathbb{Z}_p[\chi] \otimes M$ we have

$$STm = S\left(\sum_{\sigma\in\mathcal{G}} m^\sigma\right) = \varepsilon_\chi \sum_{\sigma\in\mathcal{G}} m^\sigma = \varepsilon_\chi m + \sum_{\substack{\sigma\in\mathcal{G}\\ \sigma\neq e}} \varepsilon_\chi m^\sigma = \varepsilon_\chi m = m.$$

where $\varepsilon_\chi m^\sigma = 0$ for $\sigma \neq e$ because $m^\sigma \in \varepsilon_{\chi^\sigma} M$. Since T has an inverse, the result follows. $\square$

Now we return to considering the situation where $p \mid |G|$ in general. We can define certain elements similar to idempotents in this generality.

Definition 3.11. *Let $H \subseteq G$ and let χ be a character of H with inverse $\overline{\chi}$. Define*

$$\xi_\chi = \sum_{h\in H} \overline{\chi}(h)[h].$$

ξ_χ induces a map $\xi_\chi * : M_\chi \to M^\chi$ as follows. First let $\tilde{M} = \mathbb{Z}_p[\chi] \otimes M$. Then multiplication by ξ_χ is a map $\tilde{M} \to \tilde{M}$. The kernel contains I_χ and the image is in M^χ, so this gives a map from M_χ to M^χ by Lemma 3.4.

We can understand the kernel and cokernel of ξ_χ^* by the following

Lemma 3.12. *Let $H \subseteq G$ be finite abelian groups, χ a character of H, and M a $\mathbb{Z}_p[\chi]$-module. Then we have an exact sequence of abelian groups*

$$0 \longrightarrow \hat{H}^{-1}(H, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} M) \longrightarrow M_\chi \xrightarrow{\xi_\chi^*} M^\chi \longrightarrow \hat{H}^0(H, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} \tilde{M}) \longrightarrow 0$$

Proof. We note the isomorphisms of abelian groups $M_\chi = H_0(H, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} M)$ and $M^\chi = H^0(H, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} M)$. From this we get the norm exact sequence [NSW08, Prop. 1.2.6]:

$$0 \longrightarrow \hat{H}^{-1}(H, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} M) \longrightarrow H_0(H, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} M)$$

$$\xrightarrow{N_H} H^0(H, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} M) \longrightarrow \hat{H}^0(H, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} M) \longrightarrow 0$$

Tracking through the various G actions, we see that N_H and ξ_χ^* are the same. □

4 Distributions arising from units in local fields

We now turn to a more arithmetic situation. We briefly recall some facts about the p-adic Logarithm. Let K be a p-adic local field with uniformizer π, $\mathcal{O}_K$ its ring of integers, $\mathcal{O}_K^\times$ its units, and U_1 the subgroup of units in $\mathcal{O}_K^\times$ congruent to 1 mod π. The p-adic logarithm is given by a power series and defines a Galois-equivariant homomorphism

$$\log_p : U_1 \to K.$$

The kernel of $\log_p$ is exactly the roots of unity contained in U_1, which is precisely $\mu_{p^\infty}(K)$. The logarithm can be extended to be defined on all of $K^\times$, and we call this the Iwasawa logarithm $\mathrm{Log}_p : K^\times \to K$. Its kernel is exactly the group of roots of unity and powers of p. Since its image K is p-adically complete, the Iwasawa logarithm extends to a map $\mathrm{Log}_p : \check{K}^\times \to K$, where $\check{K}^\times$ is the pro-p completion of $K^\times$[Was12, Ch. 5.1].

Now we will introduce the setting that the rest of this paper will live in. Let $F = \mathbb{Q}(\zeta_{m'})$ with $p \nmid m'$, and let $k = F(\zeta_p)$, so $k = \mathbb{Q}(\zeta_m)$ if $m = m'p$. Set $k_0 = k$, $k_n = k_0(\zeta_{p^{n+1}})$, and finally $k_\infty = \bigcup k_n$. Then k_∞/k is a cyclotomic $\mathbb{Z}_p$ extension. If we set $\Delta = \mathrm{Gal}(k/\mathbb{Q})$ and $\Gamma = \mathrm{Gal}(k_\infty/k)$, then we have $G \coloneqq \mathrm{Gal}(k_\infty/\mathbb{Q}) = \Delta \times \Gamma$.

Let $\mathcal{O}_n = \mathbb{Z}_p \otimes_\mathbb{Z} \mathcal{O}_{k_n}$. These are semilocal rings: if p splits into r primes in K_0, then it also splits into r primes in k_n. Denoting such a prime by $\mathfrak{p}_n$, then $\mathcal{O}_n \cong \mathcal{O}^r_{(k_n)_{\mathfrak{p}_n}}$. We let $K_n = \mathrm{Frac}(\mathcal{O}_n)$ be the total ring of fractions and note that $K_n = (k_n)^r_{\mathfrak{p}_n}$. Finally define

$$U_n = \{x \in \mathcal{O}_n^\times : x \equiv 1 \bmod (1 - \zeta_{p^{n+1}})\}$$
$$U_\infty = \varprojlim U_n \qquad A(K_\infty) = \varprojlim \check{K}_n,$$

where the inverse limits are taken over the norm map. The condition on U_n is equivalent to $x \equiv 1 \bmod \mathfrak{P}$ for *all* $\mathfrak{P}$ lying over p in K_n. We note an important point: if D is the decomposition group of p in Δ, then all of these semilocal objects are isomorphic to the induction of Δ over D of their purely local analogues. More precisely, let $L_n = (K_n)_{\mathfrak{p}_n}$. Then

$$\begin{aligned} K_n &= \mathrm{Ind}_D^\Delta L_n \qquad U_n = \mathrm{Ind}_D^\Delta U_1(L_n) \qquad U_\infty = \mathrm{Ind}_D^\Delta \varprojlim U_1(L_n) \\ \check{K}_n^\times &= \mathrm{Ind}_D^\Delta \check{L}_n^\times \qquad A(K_\infty) = \mathrm{Ind}_D^\Delta A(L_\infty) \qquad \mathcal{O}_n = \mathrm{Ind}_D^\Delta \mathcal{O}_{L_n}. \end{aligned} \tag{4}$$

Let γ be a topological generator of Γ such that $\gamma\zeta_{p^n} = \zeta_{p^n}^{1+m}$. This gives an isomorphism $\Gamma \cong \mathbb{Z}_p$, and fixing such an isomorphism will allow us to define the following map, that is central to our work.

Remark. The above semilocal objects, specifically $\check{K}_n^\times$ and $A(K_\infty)$ have an intepretation via étale cohomology in the same way that local analogues have an interpretation in Galois cohomology. Recall the following isomorphisms from Kummer theory

$$\check{L}_n^\times \cong H^1_{\mathrm{cts}}(L_n, \mathbb{Z}_p(1)) \qquad \text{and} \qquad A(L_\infty) \cong H^1_{\mathrm{Iw}}(L_\infty, \mathbb{Z}_p(1)) \coloneqq \varprojlim H^1_{\mathrm{cts}}(L_n, \mathbb{Z}_p(1)).$$

In the semilocal case, we obtain

$$\check{K}_n^\times \cong H^1_{\text{ét}}(K_n, \mathbb{Z}_p(1)) := \varprojlim_m H^1_{\text{ét}}(K_n, \mu_{p^m}).$$

Instead of taking an inverse limit over correstriction (which does not exist for etale cohomology) we take it over the maps induced by the map of schemes $\operatorname{Spec} K_n \to \operatorname{Spec} K_m$. It would be interesting to see how much of this could be replicated with a different Galois representation in place of $\mathbb{Z}_p(1)$, for example T_pE for an elliptic curve E.

Definition 4.1. *We define $\varrho : A(K_\infty) \to \mathcal{D}(\mathbb{Z}_p, K_\infty)$ by*

$$\varrho(x_n)(a + p^m\mathbb{Z}_p) = \mathrm{Log}_p(x_n^{\gamma^a}).$$

Here, we are taking $\mathrm{Log}_p : \check{K}_n^\times \to K_n$, and we define it on this semilocal field by using the isomorphisms in equation 4.

The distribution relations follow from the norm compatibility of (x_n) because

$$\varrho(x_n)(a+p^m\mathbb{Z}_p) = \operatorname{Log}_p(x_m^{\gamma^a}) = \operatorname{Log}_p(N_{K_m}^{K_{m+1}}(x_{m+1}^{\gamma^a})) = \sum_{b=0}^{p-1} \operatorname{Log}_p(x_{m+1}^{\gamma^{a+p^m b}}) = \sum_{b=0}^{p-1} \varrho(x_n)(a+p^m b+p^{m+1}\mathbb{Z}_p).$$

We remark that we have an isomorphism $\mathcal{D}(\mathbb{Z}_p, K_\infty) \cong \operatorname{Ind}_D^\Delta \mathcal{D}(\mathbb{Z}_p, L_\infty)$ coming from the isomorphism $K_\infty \cong \operatorname{Ind}_D^\Delta L_\infty$. The following was shown by [AW17], and follows from the behavior of ramification groups in k_∞/k.

Proposition 4.2. *The image of ϱ lies in $\mathcal{V}(\mathbb{Z}_p, K_\infty)$*

Proof. See [AW17, Thm. 3.4] □

Now that we understand the image of this map, we want to understand the domain $A(K_\infty) = \varprojlim \check{K}_n^\times$. Let N_n^∞ be the projection from $A(K_\infty)$ to $\check{K}_n^\times$, and denote its image by B_n. By properties of inverse limits, we have an isomorphism of G-modules

$$A(K_\infty) = \varprojlim_{N_{n-1}^n} B_n,$$

where the transition maps are now surjective. Let $\Gamma_n = \Gamma^{p^n}$. The Γ_n-action on B_n is trivial, so by the universal property of coinvariance there is a natural map $A(K_\infty)_{\Gamma_n} \to B_n$ which is induced by N_n^∞. This is surjective by the definition of B_n, and we claim

Proposition 4.3. *The map N_n^∞ induces an isomorphism*

$$A(K_\infty)_{\Gamma_n} \cong B_n.$$

Working coordinate-by-coordinate, we see that this is actually a special case of the following theorem (where we replace some previous notation)

Lemma 4.4. *Let K be a local field and K_∞/K be a $\mathbb{Z}_p$ extension. Let $A(K_\infty) = \varprojlim \check{K}_n^\times$, and let*

$$B_n = \bigcap_{m>n} N_n^m(\check{K}_n^\times).$$

Then $A(K_\infty)_{\Gamma_n} \cong B_n$ as Galois modules.

Proof. Let L_n be the maximal abelian p-extension of K_n, and L_∞ be the maximal abelian p-extension of L_∞. Then by class field theory we have isomorphisms of Galois modules

$$A(K_\infty) \cong \operatorname{Gal}(L_\infty/K_\infty).$$

On the Galois side, taking Γ_n-coinvariance corresponds to the extension being abelian over K_n, so we have $\operatorname{Gal}(L_\infty/K_\infty)_{\Gamma_n} = \operatorname{Gal}(L_n/K_\infty)$, which is known to correspond via class field theory to B_n. □

This gives us the corollary

Corollary 4.5. *Let $\varepsilon = 1 - \varepsilon_0$, where ε_0 is the idempotent associated to the trivial character of Π. Then*

$$\varepsilon A(K_\infty)_{\Gamma_n} \cong \varepsilon \check{K}_n^\times.$$

Proof. We can again work purely locally. Working on the Galois side and using notation from before, we have a commutative exact diagram

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{Gal}(L_n/K_\infty) \arrow[r] \arrow[d, "\sim"] & \mathrm{Gal}(L_n/K_n) \arrow[r] \arrow[d, "\sim"] & \mathrm{Gal}(K_\infty/K_n) \arrow[r] & 0 \\ & B_n \arrow[r] & \check{K}_n^\times & & \end{tikzcd}$$

Multiplying everything in site by ε and using Proposition 3.9, we obtain the isomorphism after noting that $\varepsilon\,\mathrm{Gal}(K_\infty/K_n) = 0$ since K_∞ is abelian over $\mathbb{Q}_p$. ☐

For the following discussion up through Lemma 4.8, we can work a bit more generally. Instead of starting with $\mathbb{Q}$ as the base field, we can start with an arbitrary number field, and then instead of the Cyclotomic $\mathbb{Z}_p$ extension, you can use any $\mathbb{Z}_p$ extension defined over your base field.

The Λ-module $A(K_\infty)$ is very closely related to another one, $U_\infty \cong \varprojlim U_n$. Let $\pi_i^{(n)}$ be uniformizers for the r different primes of $\mathcal{O}_n$. Then there is an isomorphism

$$\check{K}_n^\times \cong U_n \times \prod_{i=1}^{r} (\pi_i^{(n)})^{\mathbb{Z}_p^\times}$$

which follows from a similar isomorphism in the purely local case and Equation 4. However, it is important to note that this is *not* an isomorphism of Galois modules. Instead, it is better to work with a Galois-equivariant exact sequence.

Lemma 4.6. *Let all notation be as established at the beginning of this chapter. There is an exact sequence of $\Delta \times \Gamma/\Gamma^{p^n}$-modules*

$$0 \longrightarrow U_n \longrightarrow \check{K}_n^\times \xrightarrow{v} \mathrm{Ind}_D^\Delta \mathbb{Z}_p \longrightarrow 0$$

where v takes an element to its valuation at all r primes of $\mathcal{O}_n$.

Proof. We begin with the well-known exact sequence in the purely local case

$$0 \longrightarrow U_1(L_n) \longrightarrow \check{L}_n^\times \xrightarrow{v} \mathbb{Z}_p \longrightarrow 0.$$

We then apply the exact[NSW08, Prop. 1.3.6] functor Ind_D^Δ to this sequence and apply the isomorphisms in Equation 4 to obtain the lemma. ☐

Corollary 4.7. *When K_∞/K is totally ramified, there is an exact sequence of $\Lambda[\Delta]$-modules*

$$0 \longrightarrow U_\infty \longrightarrow A(K_\infty) \xrightarrow{v} \mathrm{Ind}_D^\Delta \mathbb{Z}_p \longrightarrow 0.$$

Proof. Inverse limit on compact modules is exact (see [Was12, Lem. 15.16]) so this follows from 4.6. We use the assumption on K_∞/K to verify that the norm maps are surjective on Ind_D^Δ. ☐

This essentially means that U_∞ and $A(K_\infty)$ are very close to being isomorphic, since $\mathrm{Ind}_D^\Delta \mathbb{Z}_p$ is quite small compared to these modules. It also has components corresponding to relatively few characters of Δ, which we exploit to obtain the following lemma:

Lemma 4.8. *Let χ be a character of Δ that is nontrivial on D. Then*

$$A(K_\infty)^\chi \cong U_\infty^\chi \qquad \textit{and} \qquad (\check{K}_n^\times)^\chi \cong U_n^\chi.$$

Proof. We will only handle the case on the infinite level, as the finite level case will follow by an identical argument. By Lemma 3.3, we obtain an exact sequence from Corollary 4.7

$$0 \longrightarrow U_\infty^\chi \longrightarrow A(K_\infty)^\chi \xrightarrow{v} (\mathrm{Ind}_D^\Delta \mathbb{Z}_p)^\chi.$$

Therefore it will suffice to show that $(\mathrm{Ind}_D^\Delta \mathbb{Z}_p)^\chi = 0$. By definition we have

$$(\mathrm{Ind}_D^\Delta \mathbb{Z}_p)^\chi = \mathrm{Hom}(\mathbb{Z}_p(\chi), \mathbb{Z}_p[\chi] \otimes \mathrm{Ind}_D^\Delta \mathbb{Z}_p).$$

Let f be such a homomorphism, and let $\sigma \in D$ be such that $\chi(\sigma) \neq 1$ (such a σ exists by assumption.) Then since σ acts trivially on the codomain of f

$$f(x) = \sigma f(x) = f(\sigma x) = f(\chi(\sigma)x) = \chi(\sigma) f(x)$$

so $f(x) = 0$, and $(\mathrm{Ind}_D^\Delta \mathbb{Z}_p)^\chi = 0$. ☐

Now we return to specifically considering the base field $\mathbb{Q}$ and the cyclotomic $\mathbb{Z}_p$ extension. The $\Lambda[\Delta]$-module U_∞ is well understood here thanks to Coleman theory ([Col79],[Col83]). Define $\Delta' = \mathrm{Gal}(F/\mathbb{Q})$ and $\Pi = \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$. Then we have $G = \Delta \times \Gamma = \Delta' \times \Pi \times \Gamma$. Let $\phi_p \in \Delta$ be the frobenius element of p in $F/\mathbb{Q}$.

Lemma 4.9 ([Tsu99] Th. 4.2)**.** *There is an exact sequence of $\Lambda[\Delta]$-modules*

$$0 \longrightarrow \mathrm{Ind}_D^\Delta \mathbb{Z}_p(1) \longrightarrow U_\infty \xrightarrow{Col} \mathbb{Z}_p \otimes \mathcal{O}_F[\![\Pi \times \Gamma]\!] \longrightarrow \mathbb{Z}_p \otimes \mathcal{O}_F/(1-\phi)(1) \longrightarrow 0.$$

There are frequent occurrences of $\mathbb{Z}_p \otimes \mathcal{O}_F$ motivate us to understand its Δ'-module structure. For this we can use the following general lemma:

Lemma 4.10. *Let S/R be a Galois étale extension of a DVR with Galois group G. Then $S \cong R[G]$ as an $R[G]$ module. In particular, it is induced.*

Proof. Let $\mathfrak{m}$ be the maximal ideal of R and let $R/\mathfrak{m} = \kappa$. Since S/R is Galois and étale, $S \otimes_R \kappa \cong \lambda$ is a Galois étale κ algebra. In particular, $\lambda \cong \kappa[G]$ by the normal basis theorem. Pick an $\bar{\alpha} \in \lambda$ that generates λ over $\kappa[G]$. Then have a commutative diagram

$$\begin{array}{ccc} R[G] & \xrightarrow{1\mapsto\alpha} & S \\ \downarrow & & \downarrow \\ \kappa[G] & \xrightarrow{1\mapsto\bar{\alpha}} & \lambda \end{array}$$

where α is a lift of $\bar{\alpha}$ to R. The bottom map is a surjection, so the top map is a surjection by nakayama's lemma. To verify injectivity, let π be a uniformizer of R and suppose $x\alpha = 0$ for some $x \in R[G]$. If $x \neq 0$ We may assume that $x \not\equiv 0$ by dividing by π sufficiently many times. But then $0 \equiv x\alpha \equiv \bar{x}\bar{\alpha} \bmod \mathfrak{m}$, so $\bar{x} \equiv 0 \bmod \mathfrak{m}$ since $\kappa[G] \to \lambda$ is an isomorphism. ☐

We apply this to $R = \mathbb{Z}_p$ and $S = \mathbb{Z}_p \otimes \mathcal{O}_F$ to deduce $\mathcal{O}_F \cong \mathbb{Z}_p[\Delta']$. As a result, this third term in the exact sequence in 4.9 is isomorphic to $\mathbb{Z}_p[\Delta'][\![\Pi \times \Gamma]\!] = \mathbb{Z}_p[\![G]\!]$. There is something similar we can do for the fourth term. We have

$$\mathbb{Z}_p \otimes \mathcal{O}_F/(1-\phi) \cong \mathbb{Z}_p[\Delta']/I_{\Delta' \cap D} \cong \mathrm{Ind}_{D\cap\Delta'}^{\Delta'} \mathbb{Z}_p \cong \mathrm{Ind}_D^\Delta \mathbb{Z}_p.$$

We ultimately obtain the following

Corollary 4.11. *Let all notation be as before. There is an exact sequence of $\Lambda[\Delta] = \mathbb{Z}_p[G]$-modules*

$$0 \longrightarrow \mathrm{Ind}_D^\Delta \mathbb{Z}_p(1) \longrightarrow U_\infty \longrightarrow \mathbb{Z}_p[\![G]\!] \longrightarrow \mathrm{Ind}_D^\Delta \mathbb{Z}_p(1) \longrightarrow 0.$$

Finally, we will need a result that describes the kernel of ϱ. From the above lemmas, we know that $\mathrm{Ind}_D^\Delta \mathbb{Z}_p \subseteq U_\infty \subseteq A(K_\infty)$. We show that the is exactly the kernel.

Lemma 4.12. *Suppose $x = (x_n) \in A(K_\infty)$ obeys $\varrho(x) = 0$. Then every coordinate of every x_n is a root of unity. Equivalently, x is in the copy of $\mathrm{Ind}_D^\Delta \mathbb{Z}_p$ contained in $A(K_\infty)$.*

Proof. Since ϱ works coordinate-by-coordinate, we can work purely locally, so now abuse notation and let $(x_n) \in A(L_\infty)$ such that $\mathrm{Log}_p(x_n) = 0$. Then since the kernel of Log_p is exactly $p^{\mathbb{Q}} \times \mu$, we know that $x_n = p^{a_n} \upsilon_n \in p^{\mathbb{Q}} \times \mu$ for each n. First we show that $v_p(x_n) = 0$. Supposing otherwise, then by norm compatibility there is some n such that $v_p(x_n) \in \mathbb{Q} \setminus \mathbb{Z}$. Then since $x_n, \upsilon_n \in \mathbb{Q}_p^{\mathrm{ab}}$, it follows that $x_n \upsilon_n^{-1} = p^{a_n} \in \mathbb{Q}_p^{\mathrm{ab}}$, but this is not true. Therefore $a_n = 0$ for all n, and the x_ns are roots of unity. □

5 Existence of u_χ: the semisimple case

Let all notation be as established in 4. For the duration of this section we assume that $p \nmid [F : \mathbb{Q}]$. This implies further that $p \nmid [k : \mathbb{Q}]$, which on the Galois side means $p \nmid |\Delta|$. This means we are in a *semisimple* situation. Complicated homological algebra calculations to handle representation-theoretic invariants of $\mathbb{Z}_p[\Delta]$-modules can be replaced by the simple power of Proposition 3.9.

We recall the cyclotomic units and some important results on them.

Definition 5.1. *For a cyclotomic number field $k = \mathbb{Q}(\zeta_n)$, we define the cyclotomic numbers*

$$\mathscr{C}_K^* = \langle 1 - \zeta_n^a, \zeta_n^a \rangle$$

We write $\mathscr{C}_n = \mathscr{C}_{k(\zeta_{p^{n+1}})} = \mathscr{C}_{k_n}$. Then, we may view $\mathscr{C}_n^*$ as a subgroup of $K_n^\times$. We then define $C_n^* = \bar{\mathscr{C}}_n^*$ to be the closure of $\mathscr{C}_n^*$ in $\check{K}_n^\times$. These give the *semilocoal cyclotomic units.* Because Leopoldt's conjecture is known to hold for abelian extensions of $\mathbb{Q}$, we have the isomorphism

$$C_n^* = \mathscr{C}_n^* \otimes \mathbb{Z}_p.$$

Finally, we define $C_\infty = \varprojlim C_n$, where the inverse limit is taken over the norm maps $K_{n+1}^\times \to K_n^\times$.[4]

We essentially want explicit generators of C_∞^*, however we actually only care about $(1 - \varepsilon_0)C_\infty^*$, since all of our main results will work with a character of conductor m with $p|m$ exactly once.

Lemma 5.2. *Let $r'|m'$. Define the elements $s_{r'} \in C_\infty^*$ and $\eta \in C_\infty^*$ by*

$$s_{r'} = (1 - \zeta_{rp^{n+1}})_{n \in \mathbb{N}} \qquad \textit{and} \qquad \eta = (\zeta_{p^{n+1}})_{n \in \mathbb{N}}.$$

Then (the non ε_0-parts of) $s_{r'}$ and η generate $(1 - \varepsilon_0)C_\infty^*$ over $\Lambda[\Delta]$.

Proof. Clearly, $\langle s_{r'}, \eta \rangle_{\Lambda[\Delta]} \subseteq C_\infty^*$, so by the compactness argument to show that they generate $(1 - \varepsilon_0)C_\infty^*$, we just need to show that the generate $(1 - \varepsilon_0)C_n^*$ for all n. The image of $s_{r'}$ in C_n^* is

$$(1 - \zeta_{rp^{n+1}})$$

and its Galois conjugates give $(1 - \zeta_{rp^{n+1}})^a$ for all a coprime to rp^{n+1}. Similarly, the image of η in C_n^* is ζ_{p^r}. To show that the $s_{r'}$ and η generate all of $(1 - \varepsilon_0)C_n^*$ over $\Lambda[\Delta]$, we only need to show that they span the rest of the generators of $\mathscr{C}_n^*$, or that those generators become 0 in $(1 - \varepsilon_0)C_n^*$.

For ζ_l with k not a pure power of p, we may write $\zeta_k = \zeta_{k'}^a \zeta_{p^r}^b$ for some integers a and b and k' coprime to p. But then $\zeta_{l'}$ is l'-torsion in $\mathscr{C}_n^*$, so it is killed upon passing to C_n^*, since C_n^* is a pro-p group. Therefore ζ_l is in the subgroup generated by s_r and η.

For $(1 - \zeta_r^a)$ with $p \nmid r$, this is fixed by the Galois action of Π, so therefore $\varepsilon_0(1 - \zeta_r^a) = 1 - \zeta_r^a$, so this is trivial in $(1 - \varepsilon_0)C_n^*$.

Since all generators of C_n^* are $\Lambda[\Delta]$ combinations of the projections of $s_{r'}$ and η down to C_n^*, it follows that the $s_{r'}$ and η generates C_∞^*. □

[4]Note that we are using the fact that $N_n^{n+1} C_{n+1} \subseteq C_n$, which is standard.

Now let $C_\infty = U_\infty \cap C_\infty^*$. Using Corollary 4.7 and Proposition 3.9(6) on $(1-\varepsilon_0)$, we have an isomorphism $(1-\varepsilon_0)U_\infty = (1-\varepsilon_0)A(K_\infty)$, so also $(1-\varepsilon_0)C_\infty^* \cong (1-\varepsilon_0)C_\infty$. This means we can work with U_∞ and C_∞ instead of $A(K_\infty)$ and C_∞^*.

Now we want to understand the structure of U_∞/C_∞, and here we will need to assume $p \nmid |\Delta|$. Define $\dot{T} = \frac{1+m}{1+T} - 1$. There is an isomorphism $\Lambda/\dot{T} \cong \mathbb{Z}_p(1)$ as Γ-modules. To see this, recall that $\gamma = 1+T$, so in $\Lambda/\dot{T}$ we have by rearranging $\dot{T} = 0$ that $[(1+m)\gamma^{-1}] = [1]$ or $[1+m] = [\gamma]$. Recall that we chose γ so that $\zeta_{p^r}^\gamma = \zeta_{p^n}^{1+m}$, so this equality is exactly what we need to obtain an isomorphism between $\Lambda/\dot{T}$ and $\mathbb{Z}_p(1)$. Finally, we write $\Lambda_\chi = \mathbb{Z}_p[\chi] \otimes \Lambda$.

Lemma 5.3 ([Gil79a, Gil79b]). *Let χ be an even nontrivial character of Δ, and $\varepsilon_\chi \in \mathbb{Z}_p[\chi]$ the associated idempotent. Let χ_1 be the primitive dirichlet character associated to $\chi\omega^{-1}$. Then if $\chi_1(p) \neq 1$*

$$\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty/C_\infty) \cong \Lambda_\chi/f_\chi.$$

On the other hand, if $\chi_1(p) = 1$, then there is an exact sequence

$$0 \longrightarrow \Lambda_\chi/\dot{T} \longrightarrow \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty/C_\infty) \xrightarrow{Col/\dot{T}} \Lambda_\chi/(f_\chi(T)/\dot{T}) \longrightarrow 0$$

where $G_\chi \in \mathbb{Z}_p[\chi] \otimes \Lambda$ satisfies

$$G_\chi((1+m)^{1-s} - 1) = L_p(\chi, s).$$

Now we will explain our strategy for the rest of this section. We will begin with a specific $c \in C_\infty$. Since $C_\infty \subseteq U_\infty \subseteq A(K_\infty)$, we can compute $\varrho(\xi_\chi c)$. We will show that $\varrho(\xi_\chi c) = \mu_{s,\chi}$, where we recall from Definition 2.18 that $\mu_{s,\chi}$ is a distribution on $\mathbb{Z}_p$ satisfying

$$\mu_{s,\chi}(a + p^n\mathbb{Z}_p) = \sum_{b \in \mathbb{Z}/m\mathbb{Z}} \chi(b) \log_p(1 - \zeta_{mp^n}^{\omega(b)(1+m)^a}).$$

We then use Lemma 5.3 to essentially "divide" $\xi_\chi c \in \mathbb{Z}_p[\chi] \otimes U_\infty$ by G_χ to obtain a unit $u_\chi \in \varepsilon_\chi U_\infty$, and the results of section 2.3 will imply that $\varrho(u_\chi) = \mu_{t,\chi}$. We will also show that $\xi_\chi c$ generates $\varepsilon_\chi C_\infty$, which will then imply that u_χ generates $\varepsilon_\chi U_\infty$.

Finally, by passing from K_∞ to the finite level using suitable control theorems like Proposition 4.3, we will use these equalities to give a very explicit interpretation of $L_p(1, \chi\psi)$ as factors contributing to regulators of $\varepsilon_\chi C_n$ in $\varepsilon_\chi U_n$.

Now, this is all a slight lie, as we will also have to account for the fact that we must tensor with $\mathbb{Z}_p[\chi]$ to multiply things by ε_χ, but this is a technicality that is not part of the conceptual core of the argument.

Now, we will act on this strategy.

Lemma 5.4. *Assume $p \nmid |\Delta|$ and let χ be a nontrivial even character of Δ of conductor $pr' \mid m$. Then*

$$\xi_\chi(s_{r'} \otimes 1)\Lambda_\chi = \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes C_\infty).$$

Proof. First we reduce to the case $r = m$. Let χ be a nontrivial even character and let r be its conductor. Let $U_\infty(r)$ denote the version of U_∞ we obtain by starting with $F = \mathbb{Q}(\zeta_r)$. By the Galois correspondence, we have $U_\infty(m)^{\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}(\zeta_r))} = U_\infty(r)$. Since χ is trivial on $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}(\zeta_r))$, we have $\varepsilon_\chi U_\infty(m) = \varepsilon_\chi U_\infty(m)^{\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}(\zeta_r))} = \varepsilon_\chi U_\infty(r)$.

Now suppose $t'|r'$ is a strict divisor. Since $\mathrm{Gal}(\mathbb{Q}(\zeta_{pr'})/\mathbb{Q}(\zeta_{pt'}))$ acts trivially on $(1 \otimes s_{r'})$ and the conductor of ε_χ is r, it follows that $\varepsilon_\chi(1 \otimes s_{r'}) = 0$. Finally, since $\eta \in \varepsilon_- C_\infty$, and χ is even, $\varepsilon_\chi C_\infty = \varepsilon_\chi(1 \otimes s_{r'})$. Finally, since $\xi_\chi = \text{invertible} \cdot \varepsilon_\chi$, it follows that $\xi_\chi(s_{r'} \otimes 1)$ also generates $\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes C_\infty)$. ☐

From here on out, we refer to $\xi_\chi(s_{r'} \otimes 1)$ by c_χ. With the previous lemma in mind, we now work under the assumption that $\mathrm{cond}(\chi) = m$, since $\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes C_\infty)$ is generated by c_χ independent of whether [r' is a proper divisor of m. Now we study the χ-part of ϱ. The functoriality of taking χ-parts induces a map

$$\varepsilon_\chi \varrho : \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes A(K_\infty)) \to \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes \mathcal{V}(K_\infty)).$$

Our main goal will be to study the image of U_∞ and C_∞ under this map. At least for studying the image of C_∞, it suffices to understand the image of $\xi_\chi(1 \otimes s_{m'})$, since it generates the χ-part of C_∞ by Lemma 5.4.

Lemma 5.5. *Let χ be an even character of conductor m. Then*

$$\varrho(c_\chi)(a+p^n\mathbb{Z}_p) = \sum_{b\in\mathbb{Z}/m\mathbb{Z}} \overline{\chi}(b)\otimes Log_p(1-\zeta_{mp^n}^{\omega(b)(1+m)^a}).$$

Proof. This is a computation. We have

$$\begin{aligned}\varrho(c_\chi)(a+p^n\mathbb{Z}_p) &= \xi_\chi\varrho(1\otimes s_{m'})(a+p^n\mathbb{Z}_p)\\ &= \xi_\chi(1\otimes \operatorname{Log}_p(1-\zeta_{p^{n+1}}^{(1+m)^a}))\\ &= \sum_{\delta\in\Delta}\overline{\chi}(\delta)\otimes\delta\left(\operatorname{Log}_p(1-\zeta_{mp^n}^{(1+m)^a})\right)\\ &= \sum_{b\in\mathbb{Z}/m\mathbb{Z}}\overline{\chi}(b)\otimes\operatorname{Log}_p(1-\zeta_{mp^n}^{\omega(b)(1+m)^a}).\end{aligned}$$

$\square$

Let χ be a character of conductor m, and ψ a purely wild character of conductor p^n. We define elements of $\mathbb{Z}_p[\chi]\otimes_{\mathbb{Z}_p} K_\infty$ by

$$t(\chi\psi) = \sum_{b=0}^{\operatorname{lcm}(m,p^n)-1} \chi(b)\otimes\psi(b)\zeta_{\operatorname{lcm}(m,p^n)}^b.$$

We note the following explicit relation to Gauss sums. There is an isomorphism of rings

$$C:\mathbb{Z}_p[\chi]\otimes_{\mathbb{Z}_p}K_\infty \cong \prod_{\sigma\in\operatorname{Gal}(\mathbb{Q}_p[\chi]\cap K_\infty/\mathbb{Q}_p)} K_\infty[\chi^\sigma].$$

defined by $C(a\otimes b) = (a^{\sigma_1}b, a^{\sigma_2}b,\cdots,a^{\sigma_{[\mathbb{Q}_p[\chi]\cap K_\infty:\mathbb{Q}_p]}}b)$. Then clearly we can see that

$$C(t(\chi\psi)) = (\tau(\chi^{\sigma_1}\psi),\cdots,\tau(\chi^{\sigma_{[\mathbb{Q}_p[\chi]\cap K_\infty:\mathbb{Q}_p]}}\psi)).$$

Theorem 5.6. *Let χ be a nontrivial even character, and define χ_1 to be the primitive dirichlet character associated with $\chi\omega^{-1}$. Then there exists $u_\chi\in\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty)$ such that*

a. If $\chi_1(p)\neq 1$: u_χ generates $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty)$ over Λ_χ, and

$$\varrho(u_\chi)(a+p^n\mathbb{Z}_p) = -\frac{1}{p^n}\sum_{b=0}^{p^n-1} t(\overline{\chi}\psi_{\zeta_{p^n}^a})\zeta_{p^n}^{-ab}$$

where $\chi_{\zeta_{p^n}^a}$ is the unique purely wild character that satisfies $\psi_{\zeta_{p^n}^a}(1+m)=\zeta_{p^n}^a$.

b. If $\chi_1(p)=1$: u_χ generates the image of $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty)$ under ϱ and Col, and

$$\varrho(u_\chi)(a+p^n\mathbb{Z}_p) = -\frac{1}{p^n}\sum_{b=0}^{p^n-1} t(\overline{\chi}\psi_{\zeta_{p^n}^a})(\overline{\zeta_{p^n}^a}(1+m)-1)\zeta_{p^n}^{-ab}$$

Remark. The different cases are a consequence of the different cases in Lemma 5.3.

Proof. We begin with the proof of case (a). First we show that $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty)$ is a free Λ_χ module. The conditions on χ in this case ensure that either its purely tamely ramified part is not ω^1, or it acts nontrivially on $D\subseteq\Delta$. In either case,

$$\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes\operatorname{Ind}_D^\Delta\mathbb{Z}_p(1))=0,$$

therefore by Proposition 3.9(6) and Corollary 4.11,

$$\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty)\cong\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes\mathbb{Z}_p[G_\infty])\cong\Lambda_\chi.$$

Now, recall the c_χ from Lemma 5.5. Since they generate $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes C_\infty)$, their image must be a generator of $G_\chi(T)\Lambda_\chi \subseteq \Lambda_\chi$ under the above isomorphism because of Lemma 5.3. Therefore by cyclicity of $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty)$, there is some generator u_χ of $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty)$ such that $G_\chi(T)u_\chi = c_\chi$.

It remains to show that $\varrho(u_\chi)$ has the form described. Since $G_\chi(T)u_\chi = c_\chi$, we know $G_\chi(T)\varrho(u_\chi) = \varrho(c_\chi)$. We first show that there is at most one $\mu \in \mathcal{D}(\mathbb{Z}_p, \mathbb{Z}_p[\chi]\otimes K_\infty)$ with $G_\chi\mu = \varrho(c_\chi)$. To do this, we will show that $\mathcal{D}(\mathbb{Z}_p, \mathbb{Z}_p[\chi]\otimes K_\infty)$ is $G_\chi(T)$-torsion free.

Recall the isomorphism $\mathcal{D}(\mathbb{Z}_p, \mathbb{Z}_p[\chi]\otimes K_\infty) \cong \mathcal{F}(\mu_{p^\infty}-1, \mathbb{Z}_p[\chi]\otimes K_\infty)$ from Lemma 2.11, so it suffices to show that $\mathcal{F}(\mu_{p^\infty}-1, \mathbb{Z}_p[\chi]\otimes K_\infty)$ is $G_\chi(T)$-torsion free. Suppose $0 = G_\chi(T)f(z) = G_\chi(z)f(z)$ for some $f(z) \in \mathcal{F}(\mu_{p^\infty}-1, \mathbb{Z}_p[\chi]\otimes K_\infty)$. Then if $G_\chi(\zeta_{p^n}^a - 1) \neq 0$, we have $f(\zeta_{p^n}^a - 1) = 0$ by division. But

$$G_\chi(\zeta_{p^n}^a - 1) = L_p(\chi\overline{\psi_{p_n^a}}, 1) \neq 0$$

where the first equality follows from the sentence after definition 2.20, and the second is essentially the Leopoldt conjecture for abelian number fields, see [Was12, Sec. 5.5], specifically Theorem 5.30. Therefore it follows that $f(z) = 0$ for all $z \in \mu_{p^\infty}-1$, so $f=0$ and $\mathcal{F}(\mu_{p^\infty}-1, \mathbb{Z}_p[\chi]\otimes K_\infty) \cong \mathcal{D}(\mathbb{Z}_p, \mathbb{Z}_p[\chi]\otimes K_\infty)$ is $G_\chi(T)$-torsion free.

Since there is a unique $\mu \in \mathcal{D}(\mathbb{Z}_p, \mathbb{Z}_p[\chi]\otimes K_\infty)$ with $G_\chi(T)\mu = \varrho(c_\chi)$, and $G_\chi(T)\varrho(u_\chi) = \varrho(c_\chi)$, it follows that $\varrho(u_\chi)$ is that unique μ, so we will show that

$$\mu_t(a + p_p^{\mathbb{Z}}) = -\frac{1}{p^n}\sum_{b=0}^{p^n-1} t(\overline{\chi}\psi_{\zeta_{p^n}^a})\zeta_{p^n}^{-ab} = \text{(The distribution in the theorem statement)}$$

satisfies $G_\chi(T)\mu_t = \varrho(c_\chi)$, and then $\varrho(u_\chi) = \mu_t$ will follow as argued above. Let $d = [\mathbb{Q}_p[\chi]\cap K_\infty : \mathbb{Q}_p]$, and recall the isomorphism $C : \mathbb{Z}_p[\chi]\otimes K_\infty \to K_\infty[\chi]^d$ from the discussion before the theorem statement. Then C induces an isomorphism $\mathcal{D}(\mathbb{Z}_p, \mathbb{Z}_p[\chi]\otimes K_\infty) \to \mathcal{D}(\mathbb{Z}_p, K_\infty)^d$, which we also call C. Recalling $\mu_{\tau,\chi}$ and $\mu_{s,\chi}$ from 2.3, we have

$$C(\mu_t) = (\mu_{\tau,\overline{\chi}^{\sigma_1}}, \ldots, \mu_{\tau,\overline{\chi}^{\sigma_d}}).$$

On the other hand, by Lemma 5.5 we have

$$C(\varrho(c_\chi)) = (\mu_{s,\overline{\chi}^{\sigma_1}}, \ldots, \mu_{s,\overline{\chi}^{\sigma_d}}).$$

Then, noting that G_χ acts on $K_\infty[\chi]^d$ through $(\sigma_1, \ldots, \sigma_d)$, we can compute

$$\begin{aligned}
G_\chi(T)C(\mu_t) &= G_\chi(T)(-\mu_{\tau,\overline{\chi}^{\sigma_1}}, \ldots, -\mu_{\tau,\overline{\chi}^{\sigma_d}}) \\
&= (-G_\chi^{\sigma_1}(T)\mu_{\tau,\overline{\chi}^{\sigma_1}}, \ldots, -G_\chi^{\sigma_r}(T)\mu_{\tau,\overline{\chi}^{\sigma_d}}) \\
&= (-G_{\chi_1^\sigma}(T)\mu_{\tau,\overline{\chi}^{\sigma_1}}, \ldots, -G_{\chi_r^\sigma}(T)\mu_{\tau,\overline{\chi}^{\sigma_d}}) \\
&= (-\lambda_{p,\chi^{\sigma_1}} * \mu_{\tau,\overline{\chi}^{\sigma_1}}, \ldots, -\lambda_{p,\chi^{\sigma_r}} * \mu_{\tau,\overline{\chi}^{\sigma_d}}) \\
&= (\mu_{s,\overline{\chi}^{\sigma_1}}, \ldots, \mu_{s,\overline{\chi}^{\sigma_d}}) = C(\varrho(c_\chi))
\end{aligned}$$

where $\lambda_{p,\chi}$ is the distribution on $\mathbb{Z}_p$ associated with $G_\chi(T)$, and the second to last equality follows from Theorem 2.21. The result follows.

Now we prove case (b). This will be very similar, but the fact that part of the kernel of ϱ lives in this Eigenspace will cause us minor issues. Working briefly with the χ-part of Col. Unlike before where it was trivial, we now have

$$\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes_{\mathbb{Z}_p}\operatorname{Ind}_D^\Delta \mathbb{Z}_p(1)) \cong \mathbb{Z}_p[\chi]\otimes\mathbb{Z}_p(1).$$

Therefore Proposition 3.9 and Corollary 4.11 we get an exact sequence

$$0 \longrightarrow \mathbb{Z}_p[\chi]\otimes\mathbb{Z}_p(1) \longrightarrow \varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty) \xrightarrow{\text{Col}} \Lambda_\chi \longrightarrow \mathbb{Z}_p[\chi]\otimes\mathbb{Z}_p(1) \longrightarrow 0$$

Therefore, since $\Lambda_\chi/\dot{T} \cong \mathbb{Z}_p[\chi]\otimes\mathbb{Z}_p(1)$ as Λ_χ modules, it follows that the image of Col is the ideal $(\dot{T})$.

On the other hand, the exact sequence in Lemma 5.3 tells us that the image of $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes C_\infty)$ under Col is $(G_\chi(T)) \subseteq (\dot{T})$. Therefore since c_χ generates $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes C_\infty)$, it follows that $\mathrm{Col}(c_\chi)$ is an associate of $G_\chi(T)$ in Λ_χ. Dividing $\mathrm{Col}(c_\chi)$ by $(G_\chi(T)/\dot{T})$, we obtain a generator $\kappa(T)$ of $(\dot{T}) = \mathrm{im}(\mathrm{Col})$.

Let u_χ be any preimage of $\kappa(T)$ in U_∞. We have $(G_\chi(T)/\dot{T})\mathrm{col}(u_\chi) = \mathrm{col}(c_\chi)$. Since col and ϱ have the same kernel by Lemmas 4.9 and 4.12, it follows that

$$(G_\chi(T)/\dot{T})\varrho(u_\chi) = c_\chi.$$

From here, the proof proceeds exactly like case (a), except we replace μ_t with the distribution in the statement of part (b). We replace $G_\chi(T)$ with $G_\chi(T)/\dot{T}$, $\mu_{\tau,\chi}$ with $\mu_{r,\chi}$, $\lambda_{p,\chi}$ with $\tilde{\lambda}_{p,\chi}$, and finally Theorem 2.21 with Corollary 2.23. □

We now give an application of the existence of u_χ to give an intepretation of the special value $L_p(\chi\psi, 1)$. Let χ be an even character of conductor m such that $\omega^{-1}\chi$ is also conductor m. Then we are in case (a) of Theorem 5.6.

Lemma 5.7. *We have the isomorphism*

$$\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty/C_\infty)_{\Gamma^{p^n}} = \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_n/C_n).$$

Proof. Since the conductor of χ is divisible by p, ε_χ is a multiple of $(1-\varepsilon_0)$, so we obtain $\varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_\infty)_{\Gamma_{p^n}} \cong \varepsilon_\chi(\mathbb{Z}_p[\chi]\otimes U_n)$ from the isomorphism in Corollary 4.5. The result after quotienting by C_∞ comes from noting that coinvariance is right exact and using the exact sequence

$$0 \longrightarrow \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes C_\infty) \longrightarrow \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty) \longrightarrow \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty/C_\infty) \longrightarrow 0$$

which follows from Proposition 3.9(6). □

We know the Γ^{p^n} coinvariance of $K_\infty[\![\Gamma]\!]$ is given by $K_\infty[\![\Gamma/\Gamma^{p^n}]\!]$, so therefore the coinvariance of ϱ defines a map

$$\varrho_{\Gamma^{p^n}} : \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_n) \to \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes K_\infty[\Gamma/\Gamma^{p^n}]).$$

This is explicitly given by looking at just the values of $\varrho(u)(a + p^n\mathbb{Z}_p)$, where we recall that n is now fixed.

Since χ and $\omega^{-1}\chi$ has conductor m, this map is an injection: for $u \in \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_n)$ to be in its kernel, it would need to be in the kernel of Log_p by the definition of ϱ, but this is impossible since the kernel of Log_p is contained in either the 0-space (powers of p) or 1-space (units.) Therefore, we can study $\varepsilon_\chi\mathbb{Z}_p[\chi] \otimes (U_n/C_n)$ by studying it as a subquotient of the $\mathbb{Z}_p[\chi]$-module $\mathbb{Z}_p[\chi] \otimes K_\infty[\Gamma/\Gamma^{p^n}]$.

By Lemma 5.4, Theorem 5.6, and Lemma 5.7, the image of $\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty)$ and $\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes C_\infty)$ are generated by

$$\sum_{a=0}^{p^n-1} \overline{\chi}(b) \otimes \mathrm{Log}_p(1 - \zeta_{mp^n}^{\omega(b)(1+m)^a})[\gamma^a]$$

and

$$-\sum_{a=0}^{p^n-1} \frac{1}{p^n} \sum_{b=0}^{p^n-1} t(\overline{\chi}\psi_{\zeta_{p^n}^a})\zeta_{p^n}^{-ab}.$$

Now, let ψ be a purely wild character of order p^n, and let $\zeta_\psi = \psi(1+m)$. Then we have a Fourier transform map going out of $K_\infty[\Gamma/\Gamma^{p^n}]$

$$\mathcal{F}_\psi\left(\sum_{a=0}^{p^n-1} \alpha_a[\gamma^a]\right) = \sum_{a=0}^{p^n-1} \alpha_a\overline{\psi(\gamma^a)}.$$

This is like taking a ψ-part in the following way: given $\gamma_0 \in \Gamma$, we can compute that $\mathcal{F}_\psi(\gamma_0\alpha) = \psi(\gamma_0)\mathcal{F}(\alpha)$.

We have the following commutative diagram

$$\begin{tikzcd}
\mathcal{D}(\mathbb{Z}_p, K_\infty) \cong K_\infty[\![\Gamma]\!] \arrow[d, "\text{coinvariance}"] \arrow[dr, bend left, "\mu \mapsto f_\mu(\overline{\zeta_\psi} - 1)"] & \\
K_\infty[\Gamma/\Gamma^{p^n}] \arrow[r, "\mathcal{F}_\psi"'] & K_\infty
\end{tikzcd}$$

Tensoring this entire diagram with $\mathbb{Z}_p[\chi]$, we get

$$\mathcal{F}_\psi(u_\chi) = -t(\overline{\chi\psi}) \qquad \text{and} \qquad \mathcal{F}_\psi(c_\chi) = \sum_{a=0}^{mp^n-1} \overline{\chi}(a) \otimes \overline{\psi}(a)\mathrm{Log}_p(1-\zeta_{mp^n}^a).$$

Finally, using the map $C : \mathbb{Z}_p[\chi] \otimes K_\infty \to K_\infty[\chi]$ used in the proof of 5.6 and the formula for $L_p(\chi\psi, 1)$ ([Was12, Thm 5.18]) we compute that

$$\frac{C(\mathcal{F}_\psi(c_\chi))}{C(\mathcal{F}_\psi(u_\chi))} = (L_p(\chi^{\sigma_1}\psi, 1), \ldots, L_p(\chi^{\sigma_d}\psi, 1)).$$

where $d = [\mathbb{Q}_p[\chi] \cap K_\infty : \mathbb{Q}_p]$. It follows then that the index of the module generated by $\mathcal{F}_\psi(c_\chi)$ in $\mathcal{F}_\psi(u_\chi)$ as a $\mathbb{Z}_p[\chi]$-module is

$$|L_p(\chi^{\sigma_1}\psi, 1) \cdots L_p(\chi^{\sigma_d}\psi, 1)|_p^{[\mathbb{Q}_p[\chi]:\mathbb{Q}_p[\chi]\cap K_\infty]}.$$

We can make this even nicer. Let $\theta = \sum_{\sigma \in \mathrm{Gal}(\mathbb{Q}_p[\chi]/\mathbb{Q}_p)} \chi^\sigma$. This is the character of a $[\mathbb{Q}_p[\chi] : \mathbb{Q}_p]$-dimensional $\mathbb{Q}_p$-valued representation of Δ. We know by Lemma 3.10 that $\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty/C_\infty) \cong \varepsilon_\theta U_\infty/C_\infty$, so we may think of this as the ψ part of the ϑ-part of U_∞/C_∞.

On the L-functions side, its a standard result in the theory of Artin L-functions (see [Neu99, Prop. VII.10.4(ii)]) that

$$L(\theta, s) = \prod_{\sigma \in \mathrm{Gal}(\mathbb{Q}_p[\chi]/\mathbb{Q}_p)} L(\chi^\sigma, s).$$

and this easily extends to the side of p-adic Artin L-functions. We therefore have

$$\begin{aligned} \text{Size of the } \psi\text{-part of } \varepsilon_\theta U_\infty/C_\infty &= \text{Size of the } \psi\text{-part of } \varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty/C_\infty) \\ &= |L_p(\chi^{\sigma_1}\psi, 1) \cdots L_p(\chi^{\sigma_d}\psi, 1)|_p^{[\mathbb{Q}_p[\chi]:\mathbb{Q}_p[\chi]\cap K_\infty]} \\ &= \left| \prod_{\sigma \in \mathrm{Gal}(\mathbb{Q}_p[\chi]/\mathbb{Q}_p)} L_p(\chi^\sigma \psi, 1) \right|_p \\ &= |L_p(\theta\psi, 1)|_p. \end{aligned}$$

The second to last inequality follows from the fact that $|\bullet|_p$ is Galois-invariant.

Finally, we note that the notion of ψ-part used here is not literally the notion of ψ-part that would come from directly using the ideas from section 3. That would require us to tensor with $\mathbb{Z}_p[\psi]$, and objects would become unnecessarily large very quickly, because K_∞ already contains the p^nth roots of unity. Therefore we choose to work with the more elegant $\mathcal{F}_\psi$.

We will give one more application: since the units u_χ generate $\varepsilon_\chi(U_\infty)$, their logarithms must generate the image of the logarithm map on $\varepsilon_\chi(\mathbb{Z}_p[\chi] \otimes U_\infty)$. Consider the unit

$$u = \prod_{p|\mathrm{Cond}(\chi)|m} u_\chi.$$

Proposition 5.8. *$u \in U_\infty$ and u generates $(1-\varepsilon_0)U_\infty$.*

Proof. First we show $u \in U_\infty$. If $\chi^{\sigma_1}, \cdots, \chi^{\sigma_d}$ are a complete set of Galois conjugate characters, then

$$\prod_{i=1}^d u_{\chi^{\sigma_i}} = \prod_{i=1}^d \frac{c_{\chi^{\sigma_i}}}{G_{\chi^{\sigma_i}}(T)} = N_{\mathbb{Z}_p}^{\mathbb{Z}_p[\chi]}(c_\chi/G_\chi(T)) \in U_\infty.$$

Applying this to each set of conjugates yields the result.

Now we show that u generates $(1-\varepsilon_0)U_\infty$, but this is true because $\varepsilon_\chi u$ generates the χ-component of U_∞ for all χ nontrivial on Π. ☐

It follows that u_n generates $(1-\varepsilon_0)U_n$ by 4.5, since $(1-\varepsilon_0)U_\infty = (1-\varepsilon_0)A(K_\infty)$. We now determine the image of log:

Proposition 5.9. *We have an equality of sets*

$$Log_p((1-\varepsilon_0)U_n) = \mathbb{Z}_p[\mathrm{Gal}(k_n/\mathbb{Q})]\left(\frac{1}{p^n}\sum_{p|cond(\chi)|m}\sum_{b=0}^{p^n-1} t(\overline{\chi}\psi_{p^n})\zeta_{p^n}^{-b}\right)$$

.

Proof. Since u generates $(1-\varepsilon_0)U_n$ as a Galois module, so does u^{-1}, and therefore $\mathrm{Log}_p(u^{-1}) = -\mathrm{Log}_p(u)$ generates its image under Log_p. But by definition of ϱ we have $\varrho(u)(1+p^n\mathbb{Z}_p) = \mathrm{Log}_p(u)$, so by linearity

$$\mathrm{Log}_p(u) = \sum_{p|\mathrm{cond}(\chi)|m} \varrho(u_\chi)(1+p^n\mathbb{Z}_p) = -\frac{1}{p^n}\sum_{p|\mathrm{cond}(\chi)|m}\sum_{b=0}^{p^n-1} t(\overline{\chi}\psi_{p^n})\zeta_{p^n}^{-b}.$$

□

6 Existence of u_χ: the non-semisimple case

Now we will consider the case where $p \mid |\Delta|$. While we no longer have complete semisimplicity on our side, we still have some things going for us. The theory of Idempotents applies whenever you have a subgroup of prime order, and we have two of those. The ramification group at p, $\Pi \subseteq \Delta$, is order $p-1$, and has idempotents $\varepsilon_0, \cdots, \varepsilon_{p-2} \in \mathbb{Z}_p[\Delta]$ corresponding to powers of the Teichmüller character. Similarly, the decomposition group of ∞ in Δ, given by $\{\pm 1\}$, has order 2. Since p is odd, this gives us idempotents $\varepsilon_+, \varepsilon_- \in \mathbb{Z}_p[\Delta]$.

We will of course be considering even characters, since those are the ones where the p-adic L-functions are not uniformly 0. Similarly, we will project away from the ω^0 and ω^1 eigenspaces of Π. The first because we are considering characters of conductor m, not m', and the second to avoid the problems arising in part (b) of Theorem 5.6. This means we will use the "composite idempotent"

$$\varepsilon_* = \varepsilon_+(1-\varepsilon_1)(1-\varepsilon_0).$$

The proofs in this section are essentially harder versions of the proofs in section 5. Because of this, we focus only on the new complications arising from the lack of semisimplicity. For example, the proof of Lemma 6.5 is ommited since it is exactly the same as the proof of Lemma 5.5.

Lemma 6.1. *The Coleman map gives an isomorphism*

$$\varepsilon_* U_\infty \cong \varepsilon_*\mathbb{Z}_p[\![G]\!].$$

Proof. Actually all we need to do is project away from the ω^1 eigenspace for this to be true. Using lemma 4.9 and Proposition 3.9 we have the exact sequence

$$0 \longrightarrow (1-\varepsilon_1)\,\mathrm{Ind}_D^\Delta\,\mathbb{Z}_p(1) \longrightarrow (1-\varepsilon_1)U_\infty \longrightarrow (1-\varepsilon_1)\mathbb{Z}_p[\![G]\!] \longrightarrow (1-\varepsilon_1)\,\mathrm{Ind}_D^\Delta\,\mathbb{Z}_p(1) \longrightarrow 0.$$

We note that $\varepsilon_1\mathbb{Z}_p(1) \cong \mathbb{Z}_p(1)$, so $(1-\varepsilon_1)\mathbb{Z}_p(1) = 0$. Then since $\Pi \subseteq D$, we get $(1-\varepsilon_1)\,\mathrm{Ind}_D^\Delta\,\mathbb{Z}_p(1) = 0$, so the exact sequence gives us the isomorphism $(1-\varepsilon_1)U_\infty \cong (1-\varepsilon_1)\mathbb{Z}_p[\![G]\!]$. Multiplying by $(1-\varepsilon_0)\varepsilon^+$, we obtain the result. □

In this section, we will once again need to work with the cyclotomic numbers C_∞^* and cyclotomic units C_∞, and we want to show that $\xi_\chi(1 \otimes s_{r'})$ generates C_∞^χ. However, the lack of semisimplicity will make achieving this result in general infeasible for us. We do, however, have a result for the case where $m = q^r p$ for an odd prime q, which we now begin the proof of.

Lemma 6.2. *Let $? \in \mathbb{N}$ or $? = \infty$, and let $\Lambda_? = \Lambda$ if $? = \infty$ and $\Lambda_{\Gamma^{p^n}}$ otherwise. Let s_{q^r} denote its usual definition if $? = \infty$ and its image in C_n^* if n. Then there is an exact sequence*

$$0 \longrightarrow \varepsilon_* N^{\Delta'} N^{\Gamma^z}\Lambda_?[\Delta] \longrightarrow \varepsilon_*\Lambda_?[\Delta] \xrightarrow{1\mapsto s_{q^r}} \varepsilon_* C_n^* \longrightarrow \varepsilon_*\Lambda_{\Gamma^{p^z}}[\Pi] \longrightarrow 0$$

where $z = \min\{?, v_p(q-1)\}$*,* $N^{\Delta'} = \sum_{\delta\in\Delta'}\delta$*, and* $N^{\Gamma^{p^z}} = \sum_{\gamma\in\Gamma^{p^z}}[\gamma]$ *for* $? \in \mathbb{N}$ *and* 0 *for* $n = \infty$*.*

Proof. First assume $? = n$ is finite. The result for $? = \infty$ follows from taking an inverse limit and using the fact that inverse limits are exact on compact modules [Was12, Lem. 15.16].

When n is finite, we take advantage of Bass' theorem [Was12, Thm. 8.9] that essentially classifies the Galois module structure of C_n. After tensoring the result with $\mathbb{Z}_p$ and multiplying by ε_* we see that there is an isomorphism

$$\varepsilon_*\mathbb{Z}_p \otimes A_{q^r p^n} \cong \varepsilon_* C_n$$

where $A_{q^r p^n}$ is the universal distribution on $(\mathbb{Z}/q^r p^n\mathbb{Z})$, and the map is given by $\frac{a}{q^r p^n} \mapsto 1 - \zeta_{q^r p^n}^a$. (For details on universal distributions, see [Was12, Sec. 12.3].) Then this isomorphism tells us the only relation on s_{q^r} is that $N^{\Delta'}N^{\Gamma^{p^n}} s_{q^r} = 0$, yielding exactness at $\varepsilon_*\Lambda_?[\Delta]$.

On the other hand, we see using universal distributions that $s_{q^{r'}}$ for $0 < r' < r$ is a multiple of s_q, so s_{q^r} generates all of $\varepsilon_* C_n^*/C_n^*(p)$, where $C_n^*(p)$ is the cyclotomic numbers of $\mathbb{Q}_p(\zeta_{p^{n+1}})$. On the other hand, the only elements of $C_n^*(p)$ it gets are the $\Lambda[\Delta]$ multiples of

$$\frac{1-\zeta_{p^n}}{1-\zeta_{p^n}^q} = (1-[\gamma_q])(1-\zeta_{p^n}^q)$$

where $\gamma_q(\zeta_{p^a}) = \zeta_{p^a}^q$. Since $\varepsilon_* C_n^*$ is a free $\varepsilon_*\Lambda_?[\Pi]$-module (this essentially follows from taking coinvariance of [Was12, Th. 13.55]) it follows that the cokernel is $\varepsilon_*\Lambda_?[\Pi]/(1-[\gamma_q]) \cong \varepsilon_*\Lambda_{\Gamma^{p^z}}[\Pi]$. (The Π action does not disappear since $q \equiv 1 \bmod \Pi$) ☐

Lemma 6.3. *Let* χ *be an even character of conductor* $m = q^r p$ *such that* $\chi(\Delta') \not\subseteq \mu_{p^\infty}$ *and* $\chi\omega^{-1}$ *also has conductor* m*. Then the exact sequence in Lemma 6.2 induces isomorphisms*

$$\mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_* C_\infty^*) \cong \mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_*\Lambda[\Delta]) \cong \begin{cases}\Lambda_\chi & i = 0\\ 0 & i > 0\end{cases}.$$

In particular, $(\varepsilon_* C_\infty)^\chi$ *is equal to the image of* $(\varepsilon_*\Lambda[\Delta])^\chi$*.*

Proof. The first term in the exact sequence from Lemma 6.2 vanishes for $? = \infty$, so it suffices to show by the long exact sequence in Ext and 3.9 that

$$\mathrm{Ext}^i(\mathbb{Z}_p(\chi), \Lambda_{\Gamma^{p^z}}[\Pi]) \cong \mathrm{Ext}^i(\varepsilon_*\mathbb{Z}_p(\chi), \Lambda_{\Gamma^{p^z}}[\Pi]) \cong \mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_*\Lambda_{\Gamma^{p^z}}[\Pi]).$$

vanishes for all $i \geq 0$ Using 3.7 we have a spectral sequence

$$E_2^{ij} = H^p(\Delta, \mathrm{Ext}_{\mathbb{Z}_p}(\mathbb{Z}_p(\chi), \Lambda_{\Gamma^{p^z}}[\Pi])) \Longrightarrow \mathrm{Ext}^{i+j}_{\mathbb{Z}_p[\Delta]}(\mathbb{Z}_p(\chi), \Lambda_{\Gamma^{p^z}}[\Pi]).$$

Since everything in sight is $\mathbb{Z}_p$-free, the $\mathrm{Ext}_{\mathbb{Z}_p}$ terms vanish and the spectral sequence degenerates yielding isomorphisms

$$\mathrm{Ext}^i(\mathbb{Z}_p(\chi), \Lambda_{\Gamma^{p^z}}[\Pi]) \cong H^i(\Delta, \mathrm{Hom}_{\mathbb{Z}_p}(\mathbb{Z}_p(\chi), \Lambda_{\Gamma^{p^z}})).$$

We have noncanonically $\mathrm{Hom}_{\mathbb{Z}_p}(\mathbb{Z}_p(\chi), \Lambda_{\Gamma^{p^z}})[\Pi] \cong \Lambda_{\Gamma^{p^z}}[\Pi] \otimes \mathbb{Z}_p(\chi)$. Then by Shapiro's lemma and [NSW08, 1.3.6(iii)], we obtain

$$\begin{aligned} H^i(\Delta, \mathrm{Hom}_{\mathbb{Z}_p}(\mathbb{Z}_p(\chi), \Lambda_{\Gamma^{p^z}})) &\cong H^i(\Delta, \Lambda_{\Gamma^{p^z}}[\Pi] \otimes \mathbb{Z}_p(\chi)) \\ &\cong H^i(\Delta', \Lambda_{\Gamma^{p^z}} \otimes \mathbb{Z}_p(\chi)). \end{aligned}$$

For $i = 0$, the Δ' action has no fixed points on $\Lambda_{\Gamma^{p^z}}\mathbb{Z}_p(\chi)$ because Δ' acts through the nontrivial character $\overline{\chi}|_{\Delta'}$, so H^0 vanish.

For $i > 0$, we can use periodicity: since Δ' is a cyclic group, all the H^is vanish if and only if

$$\hat{H}^0(\Delta', \Lambda_{\Gamma^{p^z}} \otimes \mathbb{Z}_p(\chi)) \cong \hat{H}^{-1}(\Delta', \Lambda_{\Gamma^{p^z}} \otimes \mathbb{Z}_p(\chi)) = 0.$$

We will show the stronger fact that $H^0(\Delta', \Lambda_{\Gamma^{p^z}} \otimes \mathbb{Z}_p(\chi))$ and $H_0(\Delta', \Lambda_{\Gamma^{p^z}} \otimes \mathbb{Z}_p(\chi))$ both vanish. We have already shown the first. For the second, let $\delta' \in \Delta'$ have $\chi(\delta') \notin \mu_{p^\infty}$. Then $1 - \overline{\chi}(\delta')$ is invertible in $\mathbb{Z}_p[\chi]$, so for any $m \in \Lambda_{\Gamma^{p^z}} \otimes \mathbb{Z}_p(\chi)$ we have

$$(1-\delta')m = (1-\overline{\chi}(\delta'))m = \text{invertible}m.$$

It follows that $\Lambda_{\Gamma^{p^z}} \otimes \mathbb{Z}_p(\chi)/I_{\Delta'}\Lambda_{\Gamma^{p^z}} \otimes \mathbb{Z}_p(\chi) = 0$. ☐

Lemma 6.4. *Let $m = q^r p$ for some $r > 0$. Let χ be an even character of conductor m such that $\chi(\Delta') \not\subseteq \mu_{p^\infty}$ and $\omega^{-1}\chi$ also has conductor m. Then*

$$C_\infty^\chi \cong \xi_\chi s_{q^r} \Lambda_\chi.$$

Proof. Since $\varepsilon_* A(K_\infty) = U_\infty$, it will suffice by 3.5 to prove this result with $\varepsilon_* C_\infty^*$ in place of C_∞. Then, by 6.3, it will suffice to show this result for $\varepsilon_*\Lambda[\Delta]$ in place of $\varepsilon_* C_\infty$, and 1 in place of s_{q^r}.

Now, since $\varepsilon_* 1$ generates $\varepsilon_*\Lambda[\Delta]$, we can use Lemma 3.12 to see that $\xi_\chi(1 \otimes \varepsilon_* s_{q^r p}) = \xi_\chi \varepsilon_* 1$ generates $\Lambda[\Delta]^\chi$ if and only if $\hat{H}^0(\Delta, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} \varepsilon_*\Lambda[\Delta])$ vanishes. Now,

$$\mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} \Lambda[\Delta] \cong \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} \varepsilon_*\Lambda[\Delta] \oplus \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} (1 - \varepsilon_*)\Lambda[\Delta]$$

so since group cohomology splits over direct sum we have

$$\begin{aligned} \hat{H}^0(\Delta, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} \varepsilon_*\Lambda[\Delta]) &\subseteq \hat{H}^0(\Delta, \mathbb{Z}_p(\chi) \otimes_{\mathbb{Z}_p} \Lambda[\Delta]) \\ &\cong \hat{H}^0(\Delta, \Lambda[\Delta]^{[\mathbb{Q}_p[\chi]:\mathbb{Q}_p]}) = 0. \end{aligned}$$

Where the isomorphism follows from [NSW08, Prop. 1.3.6(iii)]. ☐

As before, we define $c_\chi = \xi_\chi(1 \otimes s_{q^r})$.

Lemma 6.5. *We have the computation*

$$\varrho(c_\chi)(a + p^n\mathbb{Z}_p) = \sum_{b \in \mathbb{Z}/m\mathbb{Z}} \overline{\chi}(b) \otimes \mathit{Log}_p(1 - \zeta_{mp^n}^{\omega(b)(1+m)^a}).$$

Proof. This is the same as Lemma 5.5. ☐

Theorem 6.6. *The $m = pq^r$ and χ be an even character with conductor m such that $\chi(\Delta') \not\subseteq \mu_{p^\infty}$ and $\omega^{-1}\chi$ also has conductor m. Then there exists a generator u_χ of U_∞^χ such that*

$$\varrho(u_\chi) = -\frac{1}{p^n} \sum_{b=0}^{p^n-1} t(\overline{\chi}\psi_{\zeta_{p^n}^a})\zeta_{p^n}^{-ab}.$$

Proof. This proof will be analogous to the proof of Theorem 5.6. First, by taking χ parts of 6.1 and using 3.5 we get

$$\Lambda_\chi \cong \mathbb{Z}_p[\![G]\!]^\chi \cong (\varepsilon_*\mathbb{Z}_p[\![G]\!])^\chi \cong (\varepsilon_* U_\infty)^\chi \cong U_\infty^\chi.$$

We now need the following result of Tsuji

Lemma 6.7. *Let χ_1 be the primitive character associated with $\chi\omega^{-1}$. There exists a Λ_χ isomorphism*

$$U_\infty^\chi / C_\infty^\chi \cong \Lambda_\chi / G_\chi(T)$$

if and only if one of the following hold:

(i) $\chi_1(p) \notin \mu_{p^\infty}$

(ii) $\chi_1(p) \in \mu_{p^\infty}$ *and* $B_{1,\chi_1} \in \mathbb{Z}_p[\chi]^\times$

(iii) $\chi_1(p) \in \mu_{p^\infty}$ *and* $G_\chi(T)$ *has* Λ *invariant* 1.

We are in the first case because χ_1 has conductor m, a multiple of p, so we have the isomorphism. Now, by 6.4, we see that $\xi_\chi s_{q^r}$ generates C_∞^χ, so by 6.7, $\xi_\chi s_{q^r}$ generates the ideal $(G_\chi(T))$ under the isomorphism $U_\infty^\chi \cong \Lambda_\chi$, so we may divide it by $G_\chi(T)$ and obtain a unit u_χ that generates U_∞^χ. The result follows from the argument in case (a) of 5.6. ☐

Now we want to describe the analogue of the "explicit index formulas" described after the proof of Theorem 5.6. Nearly everything there extends exactly, except for 5.7. The issue is that unlike multiplying by an idempotent, taking χ-parts does not necessarily commute with taking coinvariance, so we much take this into account. Furthermore, it also does not commute with taking quotients, so we will need to pay attention to the difference between $U_\infty^\chi / C_\infty^\chi$ and $(U_\infty/C_\infty)^\chi$, at least until we prove they are the same. First we need a version of 6.3, but for finite level

Lemma 6.8. *Let χ be an even character of conductor $m = q^r p$ such that $\chi(\Delta') \not\subseteq \mu_{p^\infty}$ and $\chi\omega^{-1}$ also has conductor m. Then the exact sequence in Lemma 6.2 induces isomorphisms*

$$\mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_* C_n^*) \cong \mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_* \Lambda_{\Gamma^{p^n}}[\Delta]) \cong \begin{cases} \Lambda_{\chi,\Gamma^{p^n}} & i = 0 \\ 0 & i > 0 \end{cases}.$$

Proof. We will show the vanishing of two infinite series of Ext groups ($i \geq 0$):

$$\begin{gathered} \mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_* \Lambda_{\Gamma^{p^z}}[\Pi]) = 0 \\ \mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_* N^{\Gamma^{p^z}} N^{\Delta'} \Lambda_{\Gamma^{p^n}}[\Delta]) = 0. \end{gathered} \tag{5}$$

The vanishing of the first set was done in the proof of Lemma 6.3. For the second set, we use the isomorphism

$$\varepsilon_* N^{\Gamma^{p^z}} N^{\Delta'} \Lambda_{\Gamma^{p^n}}[\Delta] \cong \varepsilon_* \mathbb{Z}_p^{p^n}[\Pi].$$

Furthermore, we may ignore the ε_* using Proposition 3.9. We use the spectral sequence from Lemma 3.7

$$E_2^{ij} = H^i(\Delta, \mathrm{Ext}^j_{\mathbb{Z}_p}(\mathbb{Z}_p(\chi), \mathbb{Z}_p^{p^n}[\Pi])) \Longrightarrow \mathrm{Ext}^{i+j}_{\mathbb{Z}_p[\Delta]}(\mathbb{Z}_p(\chi), \mathbb{Z}_p^{p^n}[\Pi]).$$

As in Lemma 6.3, since everythng in sight is $\mathbb{Z}_p$ free, all but the 0th indexed $\mathbb{Z}_p$-Ext groups vanish, so the spectral sequence degenerates and $\mathrm{Ext}^i_{\mathbb{Z}_p[\Delta]}(\mathbb{Z}_p(\chi), \mathbb{Z}_p^{p^n}[\Pi]) \cong H^i(\Delta, \mathrm{Hom}_{\mathbb{Z}_p}(\mathbb{Z}_p(\chi), \mathbb{Z}_p^{p^n}[\Pi]))$. Then

$$\begin{aligned} H^i(\Delta, \mathrm{Hom}_{\mathbb{Z}_p}(\mathbb{Z}_p(\chi), \mathbb{Z}_p^{p^n}[\Pi])) &\cong H^i(\Delta, \mathbb{Z}_p(\overline{\chi})^{p^n}[\Pi]) \\ &\cong H^i(\Delta', \mathbb{Z}_p(\overline{\chi}^{p^n})) \end{aligned}$$

where the last isomorphism follows from Shapiro's lemma. We continue to follow the proof of 6.3 and show that $H^0(\Delta', \mathbb{Z}_p(\overline{\chi}^{p^n})) \cong H_0(\Delta', \mathbb{Z}_p(\overline{\chi}^{p^n})) = 0$, which will imply the result by periodicity of cyclic cohomology. (Recall that Δ is cyclic.)

Vanishing of H^0 is easy since $\overline{\chi}|_{\Delta'}$ is nontrivial, so since Δ' acts through $\overline{\chi}|_{\Delta'}$, it cannot have any invariant elements. For H_0, find $\delta' \in \Delta'$ with $\chi(\delta') \notin \mu_{p^\infty}$. Then $1 - \overline{\chi}(\delta')$ does not vanish, for $m \in \mathbb{Z}_p(\chi)^{p^n}$ we have

$$(1 - \delta')m = (1 - \chi(\delta'))m = (\text{invertible})m$$

so $I_{\Delta'} \mathbb{Z}_p(\chi)^{p^n} = \mathbb{Z}_p(\chi)^{p^n}$, and the coinvariance is trivial. We have now shown 5.

Now, take the exact sequence from Lemma 6.2 and split it into two short exact sequences

$$0 \longrightarrow \varepsilon_* N^\Delta N^{\Gamma^z} \Lambda_?[\Delta] \longrightarrow \varepsilon_* \Lambda_?[\Delta] \longrightarrow M \longrightarrow 0$$

$$0 \longrightarrow M \longrightarrow \varepsilon_* C_n^* \longrightarrow \varepsilon_* \Lambda_{\Gamma^z} \longrightarrow 0$$

Then taking the long exact sequence of the first we obtain $\mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_* \Lambda_?[\Delta]) \cong \mathrm{Ext}^i(\mathbb{Z}_p(\chi), M)$, and taking the long exact sequence of the second gives $\mathrm{Ext}^i(\mathbb{Z}_p(\chi), M) \cong \mathrm{Ext}^i(\mathbb{Z}_p(\chi), \varepsilon_* C_n^*)$. ☐

Lemma 6.9. *Let χ be a nontrivial even character of conductor $m = q^r p$ such that $\chi(\Delta') \not\subseteq \mu_{p^\infty}$ and $\omega^{-1}\chi$ also has conductor m. Then the canonical map*

$$U_\infty^\chi / C_\infty^\chi \to (U_\infty / C_\infty)^\chi$$

is an isomorphism, and also for any $n \geq 0$, the canonical map

$$U_n^\chi / C_n^\chi \to (U_n / C_n)^\chi$$

is an isomorphism.

Proof. Let $? = \infty$ or $? \in \mathbb{N}$, and let $\Lambda_n = \Lambda$ if $n = \infty$ or $\mathbb{Z}_p[\Gamma/\Gamma^{p^n}]$ if $n \in \mathbb{N}$. We have an exact sequence

$$0 \longrightarrow C_?^\chi \longrightarrow U_?^\chi \longrightarrow (U_?/C_?)^\chi \longrightarrow \mathrm{Ext}^1_{\mathbb{Z}_p[\Delta]}(\mathbb{Z}_p(\chi), C_?)$$

and that Ext group vanishes by Lemmas 6.3 and 6.8. ☐

Now we need to compare taking coinvariance and then χ-part to taking χ-part and then coinvariance.

Lemma 6.10. *Let $n \in \mathbb{N}$ be an integer and χ be a character of conductor $m = q^r p$ such that $\chi(\Delta') \nsubseteq \mu_{p^\infty}$ and $\omega^{-1}\chi$ also has conductor m. Then*

$$(U_\infty^\chi)_{\Gamma^{p^n}} \cong ((U_\infty)_{\Gamma^{p^n}})^\chi \cong U_n^\chi$$

and

$$(C_\infty^\chi)_{\Gamma^{p^n}} \cong ((C_\infty)_{\Gamma^{p^n}})^\chi.$$

Proof. Let F be the functor $\Lambda[\Delta] - \mathsf{Mod} \to \mathsf{Ab}$ given by taking the χ-part and the Γ^{p^n}-*invariance*. Let F_Δ denote the χ-part and F_Γ denote the Γ^{p^n}-invariance. It is easy to verify that we have a natural isomorphism of functors

$$F \simeq F_\Gamma \circ F_\Delta \simeq F_\Delta \circ F_\Gamma. \tag{6}$$

We note that $F_\Gamma = \mathrm{Hom}_\Lambda(\mathbb{Z}_p, -)$, so by [NSW08, Prop. 5.2.14] we have

$$R^i F_\Gamma \cong H^i_{\text{cts}}(\Gamma^{p^n}, -).$$

In particular, $R^1 F_\Gamma \simeq H^1_{\text{cts}}(\Gamma^{p^n}, -) = \text{Coinvariance}$. Therefore we want to show that $R^1 F_\Gamma$ and F_Δ commute for the module U_∞. By the Grothendieck spectral sequence and 6, we have two spectral sequences

$$E_2^{pq} = R^p F_\Gamma R^q F_\Delta(C_\infty) \Longrightarrow R^{p+q} F(C_\infty) \qquad \text{and} \qquad D^{pq} = R^p F_\Delta R^q F_\Gamma(C_\infty) \Longrightarrow R^{p+q} F(C_\infty).$$

For the first spectral sequence, note that (using a similar argument to the ext computation in the previous lemma)

$$R^q F_\Delta(C_\infty) \cong \mathrm{Ext}^p_{\mathbb{Z}_p(\Delta)}(\mathbb{Z}_p(\chi), U_\infty) \cong \mathrm{Ext}^p_{\mathbb{Z}_p[\Delta]}(\mathbb{Z}_p(\Delta), \varepsilon_* U_\infty) \cong \varepsilon_* \mathrm{Ext}^p_{\mathbb{Z}_p[\Delta]}(\mathbb{Z}_p(\Delta), \Lambda[\Delta]) = 0$$

therefore $E_2^{pq} = 0$ for $q > 0$, and $R^1 F(C_\infty) \cong E_2^{10} = (C_\infty^\chi)_{\Gamma^{p^n}}$. On the other hand, if $q = 0$ in the 2nd spectral sequence we have

$$R^0 F_\Gamma(U_\infty) = U_\infty^{\Gamma^{p^n}} = 0.$$

On the other hand for $q > 1$ we have

$$R^q F_\Gamma(U_\infty) \cong H^q(\Gamma^{p^n}, U_\infty) = 0$$

where the last equality follows from the cohomological dimension of $\mathbb{Z}_p$. It follows that the only nontrivial terms in D^{pq} are D^{p1}, and we obtain an isomorphism $R^1 F(U_\infty) \cong D^{01} \cong ((U_\infty)_\Gamma)^\chi$. Identifying the two expressions for $R^1 F(U_\infty)$, we obtain the first isomorphism in the lemma.

For the 2nd isomorphism, note that everything in sight commutes with multiplication by ε_*, so therefore

$$((U_\infty)_{\Gamma^{p^n}})^\chi = \varepsilon_*((U_\infty)_{\Gamma^{p^n}})^\chi = ((\varepsilon_* U_\infty)_{\Gamma^{p^n}})^\chi \cong (\varepsilon_* U_n)^\chi \cong U_n^\chi.$$

by Corollary 4.5.

For the version of the theorem involving C_∞, repeat the exact same argument, using the computation of Ext groups of C_∞ from 6.3. ☐

We now obtain the following isomorphisms

$$((U_\infty/C_\infty)^\chi)_\Gamma \cong (U_\infty^\chi/C_\infty^\chi)_\Gamma \cong ((U_\infty^\chi)_\Gamma/(C_\infty^\chi)_\Gamma) \cong (U_n^\chi/C_n^\chi) \cong (U_n/C_n)^\chi$$

by a combination of right-exactness of coinvariance, and Lemmas 6.9 and 6.

We now have enough to see that $(U_n/C_n)^\chi$ is isomorphic to the quotient of the span of u_χ by the span of c_χ in $\mathbb{Z}_p[\chi] \otimes K_\infty[\Gamma/\Gamma^{p^n}]$, and we recover the rest of section 5 as we proved it there. The one thing we miss is that the author is not aware of an analogue of 3.10 for the non semisimple case. In any case, we do get

$$\text{Size of } \psi - \text{part of } (U_n/C_n)^\chi = |L_p(\chi^{\sigma_1}\psi, 1) \cdots L_p(\chi^{\sigma_d}\psi, 1)|_p^{[\mathbb{Q}_p[\chi]:\mathbb{Q}_p[\chi]\cap K_\infty]}.$$

There is no analogue of the logarithm image results (Proposition 5.9) here, since the sum of the χ parts need not be the entire module in this case.

7 Conclusion

We finish with some directions for further work. The first and most obvious is to generalize the results of section 6. The conditions we gave were quite stringent: we needed χ to be nontrivial on Δ' and $m = q^r p$. The author believes that the main difficulties one will face are homological in nature. The more complicated the structure of C_∞, the harder it is to control the generators of the χ part.

This sort of behavior gives me the distinct impression that this theorem (the existence of u_χ) doesn't *want* to be true, it's some sort of reflection of a fact that doesn't need a bunch of Ext groups to vanish for it to be true. One potential idea here is that there is some sort of "derived" version of the existence of u_χ, however the author has no idea how one would even start on proving a fact like that.

The definitions of the Coleman map and the definition of ϱ are actually very closely related, as the key feature in each of them is a logarithm. We can see this, for example, in the fact that they have the same kernel, and they both give distributions of some kind. It would be interesting to explore a comparison between the Coleman map and ϱ.

Next, it would be interesting to see what more can be done with the fact that all the distributions involved are Volkenborn distributions. We can't quite take Amice transforms of $\varrho(u_\chi)$ and $\varrho(c_\chi)$ and divide them to obtain $G_\chi(T)$ because of the error term in Lemma 2.16, however it is true "modulo $\log_p(1+T)$." Nonetheless, this is quite a strong fact and could be useful.

Finally, it would be interesting to see what generalizations of this there are for Galois representations other than $\mathbb{Z}_p(1)$ in ways similar to whats done in [BCD$^+$14].

A Proofs on Volkenborn integration

Now it is time to subject ourselves to the proof of convergence for Volkenborn integration. Our proof differs from the ones given in [AW17] and [Bar02] since we do not use Mahler series. Instead, we take advantage of first-order taylor exapansions of differentiable functions, which exist p-adically.

The conceptual idea is this: every p-adic differentiable function is approximately an affine function $y = rx + s$ by Taylor expansion, and every Volkenborn distribution is approximately the Haar distribution. Letting $\mathbf{h}$ be the Haar distribution, we have

$$\int_{\mathbb{Z}_p} (rx+s)d\mathbf{h}(x) = \lim_{n\to\infty} \frac{1}{p^n} \sum_{a=0}^{p^n-1} rx + s = \lim_{n\to\infty} \frac{1}{p^n}\left(r\frac{p^n(p^n-1)}{2} + p^n s\right) = s - \frac{r}{2}.$$

So this Volkenborn integral converges. The idea is that all Volkenborn integrals are "close enough" to this situation to guarentee convergence.

First, we need some lemmas that say we can take Taylor expansions like this. We use the notation $B(x,r)$ for the open ball of radius r about x

Lemma A.1 (p-adic Taylor's theorem)**.** *Let $f \in S^1(\mathbb{Z}_p)$ and $x_0 \in \mathbb{Z}_p$. Then for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in B_{x_0} = B(x_0, \delta_{x_0}) \subseteq \mathbb{Z}_p$ we have*

$$f(x) = f(y) + (x-y)f'(y) + (x-y)E(x,y)$$

with $|E(x,y)|_p < \epsilon$.

Proof. By the convergence of the definition of $f'(x_0)$, we have that there is some $\delta_{x_0} > 0$ such that for all $x \in B(x, \delta_{x_0})$

$$\left| f'(y) - \frac{f(y)-f(x)}{y-x} \right|_p < \epsilon$$

From here, the form in the lemma follows from basic algebraic manipulation. ∎

Corollary A.2. *Let $f \in S^1(\mathbb{Z}_p)$ and $\varepsilon > 0$, then there exists a $\delta > 0$ such that for all $x, y \in \mathbb{Z}_p$ with $|x-y| < \delta$ we have*

$$f(x) = f(y) + (x-y)f'(y) + (x-y)E(x,y)$$

with $|E(x,y)|_p < \epsilon$ (still requiring $|x-y|_p < \delta$.)

Proof. We have an open cover $\mathcal{B} = \{B_{x_0} : x_0 \in \mathbb{Z}_p\}$, where B_{x_0} is the open set given in Lemma A.1. By compactness of $\mathbb{Z}_p$, there exists a finite subcover $\mathcal{F} = \{B_{x_1}, \ldots, B_{x_n}\} \subseteq \mathcal{B}$. Now select $\delta = \min\{\delta_{x_1}, \ldots, \delta_{x_n}\}$ and suppose $|x-y|_p < \delta$. Then since $x \in B_{x_i}$ for some i, we get $y \in B_{x_i}$ by the ultrametric inequality. The result then follows by the properties of B_{x_i} given in Lemma A.1. $\square$

Lemma A.3. *Let μ be Volkenborn and C_μ the constant from Definition 2.6. Then*

$$|\mu(a+p^n\mathbb{Z}_p)|_p \leq \max\{C_\mu, |\mu(\mathbb{Z}_p)|_p\}p^n.$$

Proof. Let $C = \max\{C_\mu, |\mu(\mathbb{Z}_p)|_p\}$. We proceed by induction on n. The $n=0$ case is just $|\mu(\mathbb{Z}_p)|_p \leq C$ which follows trivially by the definition. Then

$$|p\mu(a+p^{n+1}\mathbb{Z}_p)|_p \leq \max\{|p\mu(a+p^{n+1}\mathbb{Z}_p) - \mu(a+p^n\mathbb{Z}_p)|_p, |\mu(a+p^n\mathbb{Z}_p)|_p\} \leq \max\{C, Cp^n\} \leq Cp^n.$$

This implies $|\mu(a+p^{n+1}\mathbb{Z}_p)|_p \leq Cp^{n+1}$. $\square$

We now take the hardest step in proving Proposition 2.8, showing convergence.

Lemma A.4. *For $f \in S^1(\mathbb{Z}_p)$, the Volkenborn integral $\int_{\mathbb{Z}_p} f(x)d\mu(x)$ converges.*

Proof. This convergence is very delicate, so we must proceed carefully and take advantage of all cancellation. The basic idea is to approximate $f(x)$ linearly using Lemma A.1, and use the fact that linear functions integrate well against Volkenborn measures, as we described at the beginning of this section. Define

$$B_n = \sum_{a=0}^{p^n-1} f(a)\mu(a+p^n\mathbb{Z}_p).$$

Use Corollary A.2 to obtain a function $\epsilon(\delta)$ with $\lim_{\delta\to 0}\epsilon(\delta) = 0$ such that for all $x, y \in \mathbb{Z}_p$ with $|x-y|_p < \delta$ we have

$$f(x) = f(y) + (x-y)f'(y) + (x-y)E(x,y)$$

with $E(x,y) < \epsilon$. Moreover, we can always choose $\epsilon = p^{-2e(\delta)}$, with $e(\delta)$ going to infinity.

We will estimate $B_n - B_{n+1}$ and show that it goes to 0 as $n \to \infty$, which will imply that B_n is Cauchy by the ultrametric inequality and therefore converges. For any $\delta = p^{-k+1}$, Let n be such that $p^{-n} < \delta p^{-e(\delta)}$. We have

$$B_n - B_{n+1} = \sum_{a=0}^{p^{n-e(\delta)}-1}\sum_{b=0}^{p^{e(\delta)}-1}\Bigg(f(a+p^{n-e(\delta)}b)\mu(a+p^{n-e(\delta)}b+p^n\mathbb{Z}_p) \\ - \sum_{c=0}^{p-1} f(a+p^{n-e(\delta)}b+p^nc)\mu(a+p^{n-e(\delta)}b+p^nc+p^{n+1}\mathbb{Z}_p)\Bigg).$$

We will bound each term in the sum over a separately. Moreover, since we will only use bounds that hold uniformly over the domain $\mathbb{Z}_p$, we may assume without loss of generality that $a=0$, so we are left with bounding

$$C_n := \left|\sum_{b=0}^{p^{e(\delta)}-1}\left(f(p^{n-e(\delta)b})\mu(p^{n-e(\delta)}b+p^n\mathbb{Z}_p) - \sum_{c=0}^{p-1} f(p^{n-e(\delta)}b+p^nc)\mu(p^{n-e(\delta)}b+p^nc+p^{n+1}\mathbb{Z}_p)\right)\right|_p$$

Using Corollary A.2, we have

$$f(p^{n-e(\delta)}x) = f(0) + p^{n-e(\delta)}xf'(0) + p^{n-e(\delta)}E(x)$$

where $|E(x)| < \epsilon$ by Corollary A.2 because $|p^{n-e(\delta)}| < \delta$. Applying this estimate to C_n, we have

$$
\begin{aligned}
C_n =& \Bigg| \sum_{b=0}^{p^{e(\delta)}-1} \Bigg(\Big(f(0) + p^{n-e(\delta)} b f'(0) + p^{n-e(\delta)} b E(p^{n-e(\delta)} b) \Big) \mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) \\
& - \sum_{c=0}^{p-1} \Big(f(0) + (p^{n-e(\delta)} b + p^n c) f'(0) + (p^{n-e(\delta)} b + p^n c) E(p^{n-e(\delta)} b) \Big) \mu(p^{n-e(\delta)} b + p^n c + p^{n+1} \mathbb{Z}_p) \Bigg) \Bigg|_p \\
\leq & \Bigg| \sum_{b=0}^{p^{e(\delta)}-1} \Bigg(\Big(f(0) + p^{n-e(\delta)} b f'(0) \Big) \mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) \\
& - \sum_{c=0}^{p-1} \Big(f(0) + (p^{n-e(\delta)} b + p^n c) f'(0) \Big) \mu(p^{n-e(\delta)} b + p^n c + p^{n+1} \mathbb{Z}_p) \Bigg) \Bigg|_p \\
& + \Big| p^{n-e(\delta)} b E(p^{n-e(\delta)} b) \mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) \Big|_p \\
& + \Big| (p^{n-e(\delta)} b + p^n c) E(p^{n-e(\delta)} b + p^n c) \mu(p^{n-e(\delta)} b + p^n c + p^{n+1} \mathbb{Z}_p) \Big|_p
\end{aligned}
$$

We first analyze the error terms at the end, by Lemma A.3, we have

$$
\begin{aligned}
|\mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p)|_p &\leq M_1 p^n \\
|\mu(p^{n-e(\delta)} b + p^n c + p^{n+1} \mathbb{Z}_p)|_p &\leq M_1 p^n
\end{aligned}
$$

We similarly have the bounds

$$
|E(p^{n-e(\delta)} b)|_p \leq p^{-2e(\delta)} \qquad |E(p^{n-e(\delta)} b + p^n c)|_p \leq p^{-2e(\delta)}.
$$

so in total

$$
\begin{aligned}
& \Big| p^{n-e(\delta)} b E(p^{n-e(\delta)} b) \mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) \Big|_p \\
+ & \Big| (p^{n-e(\delta)} b + p^n c) E(p^{n-e(\delta)} b + p^n c) \mu(p^{n-e(\delta)} b + p^n c + p^{n+1} \mathbb{Z}_p) \Big|_p \leq M_2 p^{-e(\delta)}.
\end{aligned}
$$

Now we return to bounding C_n, using the fact that μ is a distribution, we may cancel the $f(0)$ terms in the sum, since

$$
\begin{aligned}
& \sum_{b=0}^{p^{e(\delta)}-1} \Bigg(f(0) \mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) - \sum_{c=0}^{p-1} f(0) \mu(p^{n-e(\delta) b + p^n c + p^{n+1} \mathbb{Z}_p}) \Bigg) \\
=& f(0) \sum_{b=0}^{p^{e(\delta)}-1} \Bigg(mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) - \sum_{c=0}^{p-1} \mu(p^{n-e(\delta) b + p^n c + p^{n+1} \mathbb{Z}_p}) \Bigg) \\
=& f(0) \sum_{b=0}^{p^{e(\delta)-1}} \Big(\mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) - \mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) \Big) = 0.
\end{aligned}
$$

Now, we are left to bound

$$
\begin{aligned}
D_n =& \Bigg| \sum_{b=0}^{p^{e(\delta)}-1} \Bigg(p^{n-e(\delta)} b f'(0) \mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) - \sum_{c=0}^{p-1} (p^{n-e(\delta)} b + p^n c f'(0)) \mu(p^{n-e(\delta)} b + p^n c + p^{n+1} \mathbb{Z}_p) \Bigg) \Bigg|_p \\
\leq & \Bigg| \sum_{b=0}^{p^{e(\delta)}-1} \Bigg(p^{n-e(\delta)} b f'(0) \mu(p^{n-e(\delta)} b + p^n \mathbb{Z}_p) - \sum_{c=0}^{p-1} p^{n-e(\delta)} b \mu(p^{n-e(\delta)} b + p^n c + p^{n+1} \mathbb{Z}_p) \Bigg) \Bigg|_p \\
+ & \Bigg| \sum_{b=0}^{p^{e(\delta)}-1} \sum_{c=0}^{p-1} p^n c \mu(p^{n-e(\delta)} b + p^n c + p^{n+1} \mathbb{Z}_p) \Bigg|_p = F_n + G_n
\end{aligned}
$$

Where we have shown that $C_n \le D_n + M_1 p^{-e(\delta)}$. Now we'll bound F_n. Since μ is Volkenborn, we have

$$\left| \mu(p^{n-e(\delta)}b + p^n c + p^{n+1}\mathbb{Z}_p) - \frac{1}{p}\mu(p^{n-e(\delta)}b + p^n\mathbb{Z}_p) \right|_p \le M_3,$$

so

$$F_n \le \left| \sum_{b=0}^{p^{e(\delta)}-1} \left(p^{n-e(\delta)} b f'(0) \mu(p^{n-e(\delta)}b + p^n\mathbb{Z}_p) - \frac{1}{p}\sum_{c=0}^{p-1} p^{n-e(\delta)} b \mu(p^{n-e(\delta)}b + p^n\mathbb{Z}_p) \right) \right| + M_3 p^{n-e(\delta)} = M_3 p^{e(\delta)-n}.$$

Now for G_n,

$$\begin{aligned}
G_n &= \left| \sum_{b=0}^{p^{e(\delta)}-1} \sum_{c=0}^{p-1} p^n c \mu(p^{n-e(\delta)}b + p^n c + p^{n+1}\mathbb{Z}_p) \right|_p \\
&\le \left| \sum_{b=0}^{p^{e(\delta)}-1} \sum_{c=0}^{p-1} p^n c \mu(p^{n-e(\delta)}b + p^n\mathbb{Z}_p) \right|_p + M_3 p^{-n} \\
&= \left| \sum_{b=0}^{p^{e(\delta)}-1} \frac{p^{n+1}(p-1)}{2} \mu(p^{n-e(\delta)}b + p^n\mathbb{Z}_p) \right|_p + M_3 p^{-n} \\
&= \left| \frac{p^{n+1}(p-1)}{2} \mu(p^{n-e(\delta)}\mathbb{Z}_p) \right|_p + M_3 p^{-n} \le M_4 p^{-e(\delta)} + M_3 p^{-n}.
\end{aligned}$$

So in total we have

$$|B_{n+1} - B_n|_p = C_n \le M_2 p^{-e(\delta)} + F_n + G_n \le M_2 p^{-e(\delta)} + M_3 p^{e(\delta)-n} + M_4 p^{-e(\delta)} + M_3 p^{-n}$$

which goes to 0 as $n, e(\delta) \to \infty$, as long as we ensure that $n > 2e(\delta)$, which we can always do (we have only required that n is large compared to $e(\delta)$, so making it larger is fine.)

Since we have show that $|B_{n+1} - B_n|_p \to 0$, the Volkenborn integral converges. □

Now we show that Volkenborn integration defines a bounded operator, the last piece of Proposition 2.8 left to prove. This will actually end up being easier than Lemma A.4, since the Ultrametric inequality means we only have to bound each term in a convergent sum to bound what they converge to.

Lemma A.5. *If μ is Volkenborn, then the linear functional $f \mapsto \int_{\mathbb{Z}_p} f(x) d\mu(x)$ is bounded. Moreover, if we denote this functional by I_μ, we have*

$$\|I_\mu\|_1 \le pC$$

where C is the constant from A.3.

Proof. Define (as in the previous proof)

$$B_n = \sum_{a=0}^{p^n-1} f(a)\mu(a + p^n\mathbb{Z}_p)$$

By the ultrametric inequality, it suffices to show

$$(1) \quad |B_0|_p \le pC\|f\|_1 \qquad (2) \quad |B_{n+1} - B_n|_p \le pC\|f\|_1$$

(1) Follows easily, since $|B_0|_p = |f(0)\mu(\mathbb{Z}_p)| \le |\mu(\mathbb{Z}_p)|_p \|f\|_1$. Now we work on (2), beginning with $|B_n - B_{n+1}|$, we have by the ultrametric inequality

$$\begin{aligned}
|B_n - B_{n+1}|_p &= \left| \sum_{a=0}^{p^n-1} \left(f(a)\mu(a + p^n\mathbb{Z}_p) - \sum_{b=0}^{p-1} f(a + p^n b)\mu(a + p^n b + p^{n+1}\mathbb{Z}_p) \right) \right|_p \\
&\le \max_a \left\{ \left| f(a)\mu(a + p^n\mathbb{Z}_p) - \sum_{b=0}^{p-1} f(a + p^n b)\mu(a + p^n b + p^{n+1}\mathbb{Z}_p) \right|_p \right\}
\end{aligned}$$

Bounding each term in the max we have

$$
\begin{aligned}
&\left|f(a)\mu(a+p^n\mathbb{Z}_p) - \sum_{b=0}^{p-1} f(a+p^nb)\mu(a+p^nb+p^{n+1}\mathbb{Z}_p)\right|_p \\
=&\left|f(a)\mu(a+p^n\mathbb{Z}_p) - \sum_{b=0}^{p-1} \left(f(a) + p^nb\frac{f(a+p^nb)-f(a)}{p^nb}\right)\mu(a+p^nb+p^{n+1}\mathbb{Z}_p)\right|_p \\
=&\left|-\sum_{b=0}^{p-1} p^nb\frac{f(a+p^nb)-f(a)}{p^nb}\mu(a+p^nb+p^{n+1}\mathbb{Z}_p)\right|_p \qquad \text{Since } \mu \text{ is a distribution} \\
\leq&\max_b\left\{\left|p^n\frac{f(a+p^nb)-f(a)}{p^nb}\mu(a+p^nb+p^{n+1}\mathbb{Z}_p)\right|_p\right\} \qquad \text{By ultrametric inequality} \\
\leq&\max_b\{p^{-n}\|f\|_1Cp^{n+1}\} \leq Cp\|f\|_1,
\end{aligned}
$$

where in the last line we are using Lemma A.3. □

This completes the proof of Proposition 2.8. Now, we'll begin to work towards Proposition 2.12. To do so, we'll need some estimates on the S^1-norm of the binomial coefficients $\binom{x}{r}$ so we can use Proposition 2.8 to pass the Volkenborm integral through a limiting operation.

Lemma A.6. *Let $r \in \mathbb{N}$, and ℓog_p denote the archimedean base-p logarithm. Then for $r > 0$*

$$\left\|\binom{x}{r}\right\|_1 = p^{\lfloor \ell og_p(r)\rfloor}$$

Proof. We see that $\binom{0}{r} = 0$, so it suffices to bound

$$\sup_{x\neq y}\left\|\frac{\binom{x}{r}-\binom{y}{r}}{x-y}\right\|_p.$$

We use the well-known identity

$$\sum_{i+j=r}\binom{a}{i}\binom{b}{j} = \binom{a+b}{r}$$

with $a = x-y$ and $b = y$. We then compute

$$\binom{x}{r} - \binom{y}{r} = \sum_{i+j=r}\binom{x-y}{i}\binom{y}{j} - \binom{y}{j} = \sum_{i=1}^{r}\binom{x-y}{i}\binom{y}{r-j}$$

Using Kummer's theorem, we see that $\nu_p(\binom{x-y}{i}) \geq \nu_p(x-y) - \lfloor \ell\text{og}_p(r)\rfloor$, so using the triangle inequality we have the bound

$$\left\|\frac{\binom{x}{r}-\binom{y}{r}}{x-y}\right\|_p \leq p^{\lfloor \ell\text{og}_p(r)\rfloor}.$$

□

Now

Proof of Proposition 2.12. Let $z \in \mathbb{C}_p$ with $|z|_p < 1$ and $x \in \mathbb{Z}_p$ be fixed. By definition

$$(1+z)^x = \sum_{n=0}^{\infty}\binom{x}{r}z^r.$$

We want to show that this holds in the S^1 norm when we vary x. Letting $x \in \mathbb{Z}_p$ be arbitrary now, we have

$$\left\| \binom{x}{r} z^r \right\|_1 = |z^r|_p \left\| \binom{x}{r} \right\|_1 \leq |z|^r p^{\lfloor \ell\text{og}_p r \rfloor},$$

since $|z| < 1$, this goes to 0, and so

$$\sum_{n=0}^{\infty} \binom{x}{r} z^r$$

converges in the S^1 norm by the ultrametric inequality. It therefore must converge to $(1+z)^x$ by uniqueness. For the Volkenborn measure μ from the proposition statement, let I_μ be the "integration against μ" functional. It is bounded by Proposition 2.8, so therefore we may pass it through limits and

$$\int_{\mathbb{Z}_p} (1+z)^x d\mu(x) = I_\mu (1+z)^x = I_\mu \lim_{n\to\infty} \sum_{r=0}^{n} z^r \binom{x}{r} = \lim_{n\to\infty} \sum_{r=0}^{n} z^r I_\mu \binom{x}{r} = \sum_{r=0}^{\infty} \left(\int_{\mathbb{Z}_p} \binom{x}{r} d\mu(x) \right) z^r.$$

□

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